

5.8 The saturation concentration of dissolved oxygen in freshwater can be calculated with the equation

$$\ln o_{sf} = -139.34411 + \frac{1.575701 \times 10^5}{T_a} - \frac{6.642308 \times 10^7}{T_a^2} + \frac{1.243800 \times 10^{10}}{T_a^3} - \frac{8.621949 \times 10^{11}}{T_a^4}$$

where o_{sf} = the saturation concentration of dissolved oxygen in freshwater at 1 atm (mg L⁻¹); and T_a = absolute temperature (K). Remember that $T_a = T + 273.15$, where T = temperature (°C). According to this equation, saturation decreases with increasing temperature. For typical natural waters in temperate climates, the equation can be used to determine that oxygen concentration ranges from 14.621 mg/L at 0 °C to 6.949 mg/L at 35 °C. Given a value of oxygen concentration, this formula and the bisection method can be used to solve for temperature in °C.

- (a) If the initial guesses are set as 0 and 35 °C, how many bisection iterations would be required to determine temperature to an absolute error of 0.05 °C?
- (b) Based on (a), develop and test a bisection M-file function to determine T as a function of a given oxygen concentration. Test your function for $o_{sf} = 8, 10$ and 14 mg/L. Check your results.

5.9 A beam is loaded as shown in Fig. P5.9. Use the bisection method to solve for the position inside the beam where there is no moment.

5.10 Water is flowing in a trapezoidal channel at a rate of $Q = 20$ m³/s. The critical depth y for such a channel must satisfy the equation

$$0 = 1 - \frac{Q^2}{gA_c^3}B$$

where $g = 9.81$ m/s², A_c = the cross-sectional area (m²), and B = the width of the channel at the surface (m). For this case, the width and the cross-sectional area can be related to depth y by

$$B = 3 + y$$

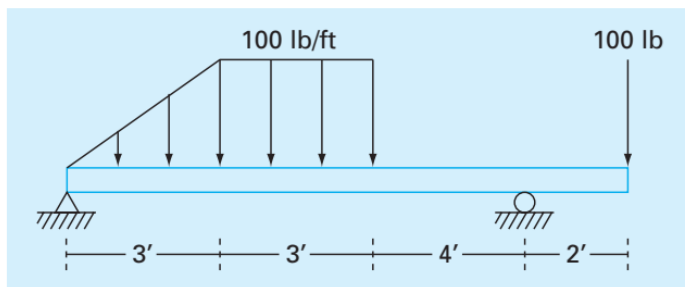


FIGURE P5.9

and

$$A_c = 3y + \frac{y^2}{2}$$

Solve for the critical depth using (a) the graphical method, (b) bisection, and (c) false position. For (b) and (c) use initial guesses of $x_l = 0.5$ and $x_u = 2.5$, and iterate until the approximate error falls below 1% or the number of iterations exceeds 10. Discuss your results.

5.11 The Michaelis-Menten model describes the kinetics of enzyme mediated reactions:

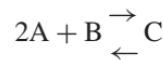
$$\frac{dS}{dt} = -v_m \frac{S}{k_s + S}$$

where S = substrate concentration (moles/L), v_m = maximum uptake rate (moles/L/d), and k_s = the half-saturation constant, which is the substrate level at which uptake is half of the maximum [moles/L]. If the initial substrate level at $t = 0$ is S_0 , this differential equation can be solved for

$$S = S_0 - v_m t + k_s \ln(S_0/S)$$

Develop an M-file to generate a plot of S versus t for the case where $S_0 = 8$ moles/L, $v_m = 0.7$ moles/L/d, and $k_s = 2.5$ moles/L.

5.12 A reversible chemical reaction



can be characterized by the equilibrium relationship

$$K = \frac{c_c}{c_a^2 c_b}$$

where the nomenclature c_i represents the concentration of constituent i . Suppose that we define a variable x as representing the number of moles of C that are produced. Conservation of mass can be used to reformulate the equilibrium relationship as

$$K = \frac{(c_{c,0} + x)}{(c_{a,0} - 2x)^2 (c_{b,0} - x)}$$

where the subscript 0 designates the initial concentration of each constituent. If $K = 0.016$, $c_{a,0} = 42$, $c_{b,0} = 28$, and $c_{c,0} = 4$, determine the value of x .

- (a) Obtain the solution graphically.
- (b) On the basis of (a), solve for the root with initial guesses of $x_l = 0$ and $x_u = 20$ to $\varepsilon_s = 0.5\%$. Choose either bisection or false position to obtain your solution. Justify your choice.

5.13 Figure P5.13a shows a uniform beam subject to a linearly increasing distributed load. The equation for the resulting elastic curve is (see Fig. P5.13b)

$$y = \frac{w_0}{120EI L} (-x^5 + 2L^2 x^3 - L^4 x) \quad (\text{P5.13})$$

Use bisection to determine the point of maximum deflection (i.e., the value of x where $dy/dx = 0$). Then substitute this value into Eq. (P5.13) to determine the value of the maximum deflection. Use the following parameter values in your

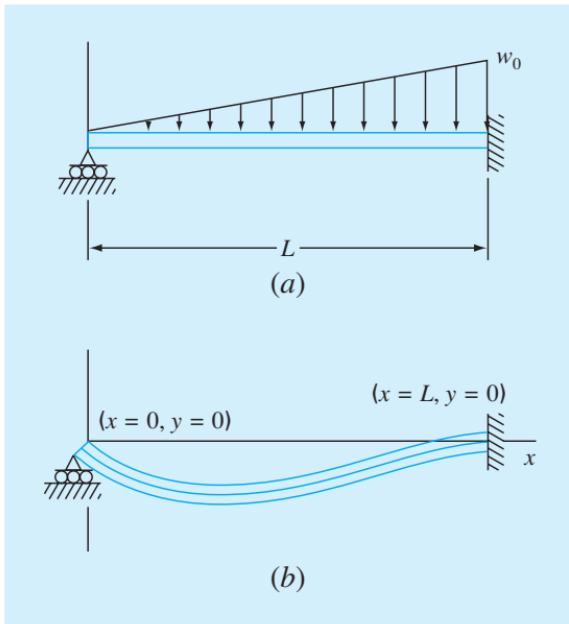


FIGURE P5.13

computation: $L = 600$ cm, $E = 50,000$ kN/cm², $I = 30,000$ cm⁴, and $w_0 = 2.5$ kN/cm.

5.14 You buy a \$35,000 vehicle for nothing down at \$8,500 per year for 7 years. Use the `bisect` function from Fig. 5.7 to determine the interest rate that you are paying. Employ initial guesses for the interest rate of 0.01 and 0.3 and a stopping criterion of 0.00005. The formula relating present worth P , annual payments A , number of years n , and interest rate i is

$$A = P \frac{i(1+i)^n}{(1+i)^n - 1}$$

5.15 Many fields of engineering require accurate population estimates. For example, transportation engineers might find it necessary to determine separately the population growth trends of a city and adjacent suburb. The population of the urban area is declining with time according to

$$P_u(t) = P_{u,\max} e^{-k_u t} + P_{u,\min}$$

while the suburban population is growing, as in

$$P_s(t) = \frac{P_{s,\max}}{1 + [P_{s,\max}/P_0 - 1]e^{-k_s t}}$$

where $P_{u,\max}$, k_u , $P_{s,\max}$, P_0 , and k_s = empirically derived parameters. Determine the time and corresponding values of $P_u(t)$ and $P_s(t)$ when the suburbs are 20% larger than the city. The parameter values are $P_{u,\max} = 80,000$, $k_u = 0.05/\text{yr}$, $P_{u,\min} = 110,000$ people, $P_{s,\max} = 320,000$ people, $P_0 = 10,000$ people, and $k_s = 0.09/\text{yr}$. To obtain your solutions, use (a) graphical, and (b) false-position methods.

5.16 The resistivity ρ of doped silicon is based on the charge q on an electron, the electron density n , and the electron mobility μ . The electron density is given in terms of the doping density N and the intrinsic carrier density n_i . The electron mobility is described by the temperature T , the reference temperature T_0 , and the reference mobility μ_0 . The equations required to compute the resistivity are

$$\rho = \frac{1}{qn\mu}$$

where

$$n = \frac{1}{2} \left(N + \sqrt{N^2 + 4n_i^2} \right) \quad \text{and} \quad \mu = \mu_0 \left(\frac{T}{T_0} \right)^{-2.42}$$

Determine N , given $T_0 = 300$ K, $T = 1000$ K, $\mu_0 = 1360$ cm² (V s)⁻¹, $q = 1.7 \times 10^{-19}$ C, $n_i = 6.21 \times 10^9$ cm⁻³, and a desired $\rho = 6.5 \times 10^6$ V s cm/C. Employ initial guesses of $N = 0$ and 2.5×10^{10} . Use (a) bisection and (b) the false position method.

5.17 A total charge Q is uniformly distributed around a ring-shaped conductor with radius a . A charge q is located at a distance x from the center of the ring (Fig. P5.17). The force exerted on the charge by the ring is given by

$$F = \frac{1}{4\pi\epsilon_0} \frac{qQx}{(x^2 + a^2)^{3/2}}$$

where $\epsilon_0 = 8.9 \times 10^{-12}$ C²/(N m²). Find the distance x where the force is 1.25 N if q and Q are 2×10^{-5} C for a ring with a radius of 0.85 m.

5.18 For fluid flow in pipes, friction is described by a dimensionless number, the *Fanning friction factor* f . The Fanning friction factor is dependent on a number of parameters related to the size of the pipe and the fluid, which can all be represented by another dimensionless quantity, the *Reynolds number* Re . A formula that predicts f given Re is the *von Karman equation*:

$$\frac{1}{\sqrt{f}} = 4 \log_{10} (Re\sqrt{f}) - 0.4$$

Typical values for the Reynolds number for turbulent flow are 10,000 to 500,000 and for the Fanning friction factor are 0.001 to 0.01. Develop a function that uses bisection to solve for f given a user-supplied value of Re between 2,500 and

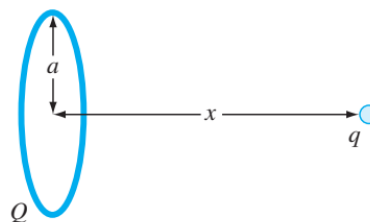


FIGURE P5.17

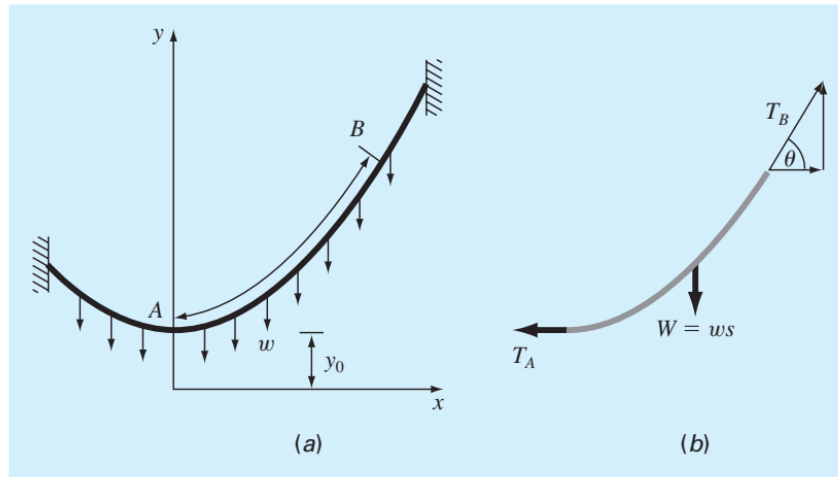


FIGURE P6.17

where R = the universal gas constant [$= 0.518 \text{ kJ}/(\text{kg K})$], T = absolute temperature (K), p = absolute pressure (kPa), and v = the volume of a kg of gas (m^3/kg). The parameters a and b are calculated by

$$a = 0.427 \frac{R^2 T_c^{2.5}}{p_c} \quad b = 0.0866 R \frac{T_c}{p_c}$$

where $p_c = 4600 \text{ kPa}$ and $T_c = 191 \text{ K}$. As a chemical engineer, you are asked to determine the amount of methane fuel that can be held in a 3-m^3 tank at a temperature of -40°C with a pressure of $65,000 \text{ kPa}$. Use a root-locating method of your choice to calculate v and then determine the mass of methane contained in the tank.

6.16 The volume of liquid V in a hollow horizontal cylinder of radius r and length L is related to the depth of the liquid h by

$$V = \left[r^2 \cos^{-1} \left(\frac{r-h}{r} \right) - (r-h) \sqrt{2rh - h^2} \right] L$$

Determine h given $r = 2 \text{ m}$, $L = 5 \text{ m}^3$, and $V = 8 \text{ m}^3$.

6.17 A catenary cable is one which is hung between two points not in the same vertical line. As depicted in Fig. P6.17a, it is subject to no loads other than its own weight. Thus, its weight acts as a uniform load per unit length along the cable w (N/m). A free-body diagram of a section AB is depicted in Fig. P6.17b, where T_A and T_B are the tension forces at the end. Based on horizontal and vertical force balances, the following differential equation model of the cable can be derived:

$$\frac{d^2 y}{dx^2} = \frac{w}{T_A} \sqrt{1 + \left(\frac{dy}{dx} \right)^2}$$

Calculus can be employed to solve this equation for the height of the cable y as a function of distance x :

$$y = \frac{T_A}{w} \cosh \left(\frac{w}{T_A} x \right) + y_0 - \frac{T_A}{w}$$

(a) Use a numerical method to calculate a value for the parameter T_A given values for the parameters $w = 10$ and $y_0 = 5$, such that the cable has a height of $y = 15$ at $x = 50$.

(b) Develop a plot of y versus x for $x = -50$ to 100 .

6.18 An oscillating current in an electric circuit is described by $I = 9e^{-t} \sin(2\pi t)$, where t is in seconds. Determine all values of t such that $I = 3.5$.

6.19 Figure P6.19 shows a circuit with a resistor, an inductor, and a capacitor in parallel. Kirchhoff's rules can be used to express the impedance of the system as

$$\frac{1}{Z} = \sqrt{\frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L} \right)^2}$$

where Z = impedance (Ω), and ω is the angular frequency. Find the ω that results in an impedance of 100Ω using the `fzero` function with initial guesses of 1 and 1000 for the following parameters: $R = 225 \Omega$, $C = 0.6 \times 10^{-6} \text{ F}$, and $L = 0.5 \text{ H}$.

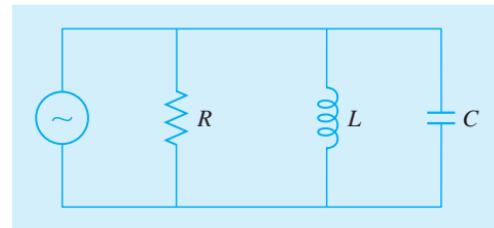


FIGURE P6.19

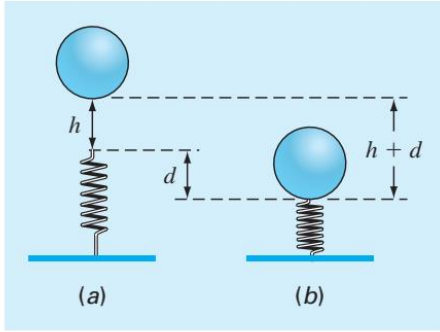


FIGURE P6.20

6.20 Real mechanical systems may involve the deflection of nonlinear springs. In Fig. P6.20, a block of mass m is released a distance h above a nonlinear spring. The resistance force F of the spring is given by

$$F = -(k_1 d + k_2 d^{3/2})$$

Conservation of energy can be used to show that

$$0 = \frac{2k_2 d^{5/2}}{5} + \frac{1}{2}k_1 d^2 - mgd - mgh$$

Solve for d , given the following parameter values: $k_1 = 40,000 \text{ g/s}^2$, $k_2 = 40 \text{ g/(s}^2 \text{ m}^5)$, $m = 95 \text{ g}$, $g = 9.81 \text{ m/s}^2$, and $h = 0.43 \text{ m}$.

6.21 Aerospace engineers sometimes compute the trajectories of projectiles such as rockets. A related problem deals with the trajectory of a thrown ball. The trajectory of a ball thrown by a right fielder is defined by the (x, y) coordinates as displayed in Fig. P6.21. The trajectory can be modeled as

$$y = (\tan \theta_0)x - \frac{g}{2v_0^2 \cos^2 \theta_0}x^2 + y_0$$

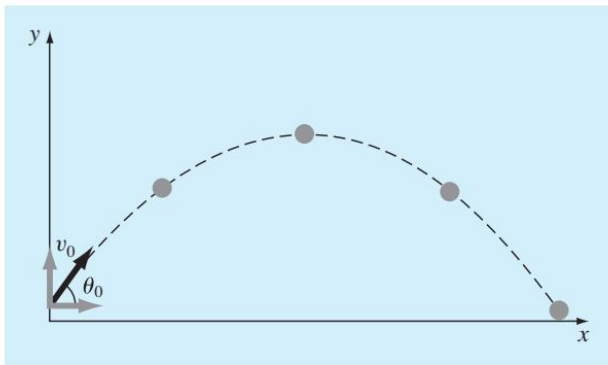


FIGURE P6.21

Find the appropriate initial angle θ_0 , if $v_0 = 30 \text{ m/s}$, and the distance to the catcher is 90 m . Note that the throw leaves the right fielder's hand at an elevation of 1.8 m and the catcher receives it at 1 m .

6.22 You are designing a spherical tank (Fig. P6.22) to hold water for a small village in a developing country. The volume of liquid it can hold can be computed as

$$V = \pi h^2 \frac{[3R - h]}{3}$$

where V = volume [m^3], h = depth of water in tank [m], and R = the tank radius [m].

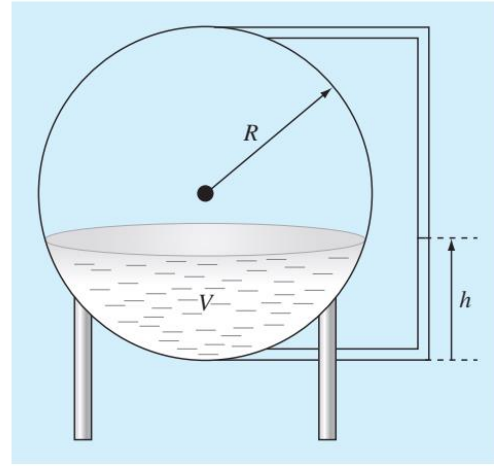


FIGURE P6.22

If $R = 3 \text{ m}$, what depth must the tank be filled to so that it holds 30 m^3 ? Use three iterations of the most efficient numerical method possible to determine your answer. Determine the approximate relative error after each iteration. Also, provide justification for your choice of method. Extra information: **(a)** For bracketing methods, initial guesses of 0 and R will bracket a single root for this example. **(b)** For open methods, an initial guess of R will always converge.

6.23 Perform the identical MATLAB operations as those in Example 6.8 to manipulate and find all the roots of the polynomial

$$f_5(x) = (x + 2)(x + 5)(x - 6)(x - 4)(x - 8)$$

6.24 In control systems analysis, transfer functions are developed that mathematically relate the dynamics of a system's input to its output. A transfer function for a robotic positioning system is given by

$$G(s) = \frac{C(s)}{N(s)} = \frac{s^3 + 9s^2 + 26s + 24}{s^4 + 15s^3 + 77s^2 + 153s + 90}$$

where $G(s)$ = system gain, $C(s)$ = system output, $N(s)$ = system input, and s = Laplace transform complex frequency. Use MATLAB to find the roots of the numerator and denominator and factor these into the form

$$G(s) = \frac{(s + a_1)(s + a_2)(s + a_3)}{(s + b_1)(s + b_2)(s + b_3)(s + b_4)}$$

where a_i and b_i = the roots of the numerator and denominator, respectively.

6.25 The Manning equation can be written for a rectangular open channel as

$$Q = \frac{\sqrt{S}(BH)^{5/3}}{n(B + 2H)^{2/3}}$$

where Q = flow (m^3/s), S = slope (m/m), H = depth (m), and n = the Manning roughness coefficient. Develop a fixed-point iteration scheme to solve this equation for H given $Q = 5$, $S = 0.0002$, $B = 20$, and $n = 0.03$. Perform the computation until ε_a is less than $\varepsilon_s = 0.05\%$. Prove that your scheme converges for all initial guesses greater than or equal to zero.

6.26 See if you can develop a foolproof function to compute the friction factor based on the Colebrook equation as described in Sec. 6.7. Your function should return a precise result for Reynolds number ranging from 4000 to 10^7 and for ε/D ranging from 0.00001 to 0.05.

6.27 Use the Newton-Raphson method to find the root of

$$f(x) = e^{-0.5x}(4 - x) - 2$$

Employ initial guesses of (a) 2, (b) 6, and (c) 8. Explain your results.

6.28 Given

$$f(x) = -2x^6 - 1.5x^4 + 10x + 2$$

Use a root-location technique to determine the maximum of this function. Perform iterations until the approximate relative error falls below 5%. If you use a bracketing method, use initial guesses of $x_l = 0$ and $x_u = 1$. If you use the Newton-Raphson or the modified secant method, use an initial guess of $x_i = 1$. If you use the secant method, use initial

guesses of $x_{i-1} = 0$ and $x_i = 1$. Assuming that convergence is not an issue, choose the technique that is best suited to this problem. Justify your choice.

6.29 You must determine the root of the following easily differentiable function:

$$e^{0.5x} = 5 - 5x$$

Pick the best numerical technique, justify your choice, and then use that technique to determine the root. Note that it is known that for positive initial guesses, all techniques except fixed-point iteration will eventually converge. Perform iterations until the approximate relative error falls below 2%. If you use a bracketing method, use initial guesses of $x_l = 0$ and $x_u = 2$. If you use the Newton-Raphson or the modified secant method, use an initial guess of $x_i = 0.7$. If you use the secant method, use initial guesses of $x_{i-1} = 0$ and $x_i = 2$.

6.30 (a) Develop an M-file function to implement Brent's root-location method. Base your function on Fig. 6.10, but with the beginning of the function changed to

```
function [b,fb] = fzeronew(f,xl,xu,varargin)
% fzeronew: Brent root location zeroes
% [b,fb] = fzeronew(f,xl,xu,p1,p2,...):
%   uses Brent's method to find the root of f
% input:
%   f = name of function
%   xl, xu = lower and upper guesses
%   p1,p2,... = additional parameters used by f
% output:
%   b = real root
%   fb = function value at root
```

Make the appropriate modifications so that the function performs as outlined in the documentation statements. In addition, include error traps to ensure that the function's three required arguments (f, xl, xu) are prescribed, and that the initial guesses bracket a root.

(b) Test your function by using it to solve for the root of the function from Example 5.6 using

```
>> [x,fx] = fzeronew(@(x,n) x^n-1,0,1.3,10)
```

PROBLEMS

8.1 Given a square matrix $[A]$, write a single line MATLAB command that will create a new matrix $[Aug]$ that consists of the original matrix $[A]$ augmented by an identity matrix $[I]$.

8.2 A number of matrices are defined as

$$[A] = \begin{bmatrix} 4 & 5 \\ 1 & 2 \\ 5 & 6 \end{bmatrix} \quad [B] = \begin{bmatrix} 4 & 3 & 7 \\ 1 & 2 & 6 \\ 2 & 0 & 4 \end{bmatrix}$$

$$\{C\} = \begin{Bmatrix} 2 \\ 6 \\ 1 \end{Bmatrix} \quad [D] = \begin{bmatrix} 5 & 4 & 3 & -7 \\ 2 & 1 & 7 & 5 \end{bmatrix}$$

$$[E] = \begin{bmatrix} 1 & 5 & 6 \\ 7 & 1 & 3 \\ 4 & 0 & 6 \end{bmatrix}$$

$$[F] = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 7 & 4 \end{bmatrix} \quad [G] = \begin{bmatrix} 8 & 6 & 4 \end{bmatrix}$$

Answer the following questions regarding these matrices:

- What are the dimensions of the matrices?
- Identify the square, column, and row matrices.
- What are the values of the elements: a_{12} , b_{23} , d_{32} , e_{22} , f_{12} , g_{12} ?
- Perform the following operations:

$$(1) [E] + [B] \quad (2) [A] + [F] \quad (3) [B] - [E]$$

$$(4) 7 \times [B] \quad (5) \{C\}^T \quad (6) [E] \times [B]$$

$$(7) [B] \times [E] \quad (8) [D]^T \quad (9) [G] \times \{C\}$$

$$(10) [I] \times [B] \quad (11) [E]^T \times [E] \quad (12) \{C\}^T \times \{C\}$$

8.3 Write the following set of equations in matrix form:

$$50 = 5x_3 - 6x_2$$

$$2x_2 + 7x_3 + 30 = 0$$

$$x_1 - 7x_3 = 50 - 3x_2 + 5x_1$$

Use MATLAB to solve for the unknowns. In addition, use it to compute the transpose and the inverse of the coefficient matrix.

8.4 Three matrices are defined as

$$[A] = \begin{bmatrix} 6 & -1 \\ 12 & 7 \\ -5 & 3 \end{bmatrix} \quad [B] = \begin{bmatrix} 4 & 0 \\ 0.6 & 8 \end{bmatrix} \quad [C] = \begin{bmatrix} 1 & -2 \\ -6 & 1 \end{bmatrix}$$

- Perform all possible multiplications that can be computed between pairs of these matrices.
- Justify why the remaining pairs cannot be multiplied.

(c) Use the results of (a) to illustrate why the order of multiplication is important.

8.5 Solve the following system with MATLAB:

$$\begin{bmatrix} 3+2i & 4 \\ -i & 1 \end{bmatrix} \begin{Bmatrix} z_1 \\ z_2 \end{Bmatrix} = \begin{Bmatrix} 2+i \\ 3 \end{Bmatrix}$$

8.6 Develop, debug, and test your own M-file to multiply two matrices—that is, $[X] = [Y][Z]$, where $[Y]$ is m by n and $[Z]$ is n by p . Employ `for...end` loops to implement the multiplication and include error traps to flag bad cases. Test the program using the matrices from Prob. 8.4.

8.7 Develop, debug, and test your own M-file to generate the transpose of a matrix. Employ `for...end` loops to implement the transpose. Test it on the matrices from Prob. 8.4.

8.8 Develop, debug, and test your own M-file function to switch the rows of a matrix using a permutation matrix. The first lines of the function should be as follows:

```
function B = permut(A,r1,r2)
% Permut: Switch rows of matrix A
% with a permutation matrix
% B = permut(A,r1,r2)
% input:
% A = original matrix
% r1, r2 = rows to be switched
% output:
% B = matrix with rows switched
```

Include error traps for erroneous inputs (e.g., user specifies rows that exceed the dimensions of the original matrix).

8.9 Five reactors linked by pipes are shown in Fig. P8.9. The rate of mass flow through each pipe is computed as the product of flow (Q) and concentration (c). At steady state, the mass flow into and out of each reactor must be equal. For example, for the first reactor, a *mass balance* can be written as

$$Q_{01}c_{01} + Q_{31}c_3 = Q_{15}c_1 + Q_{12}c_1$$

Write mass balances for the remaining reactors in Fig. P8.9 and express the equations in matrix form. Then use MATLAB to solve for the concentrations in each reactor.

8.10 An important problem in structural engineering is that of finding the forces in a statically determinate truss (Fig. P8.10). This type of structure can be described as a system of coupled linear algebraic equations derived from force balances. The sum of the forces in both horizontal and

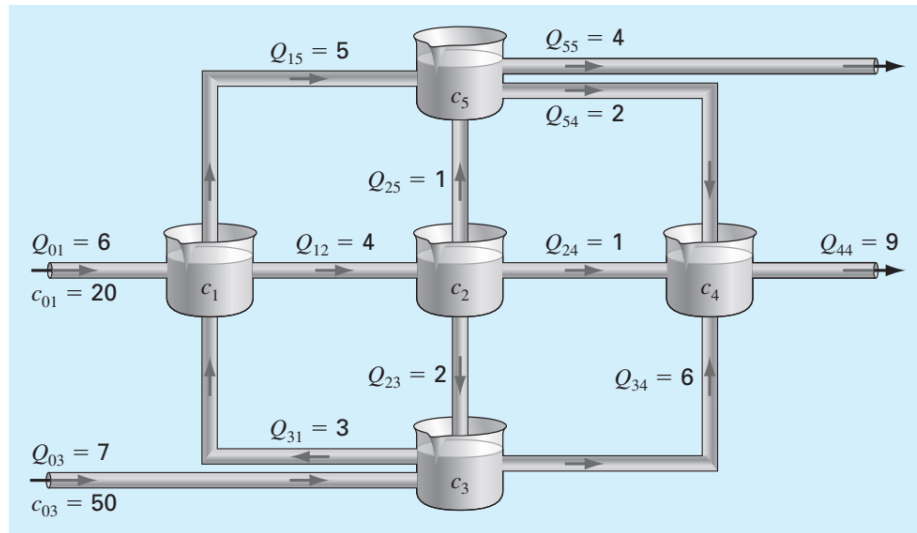


FIGURE P8.9

vertical directions must be zero at each node, because the system is at rest. Therefore, for node 1:

$$\begin{aligned}\sum F_H &= 0 = -F_1 \cos 30^\circ + F_3 \cos 60^\circ + F_{1,h} \\ \sum F_V &= 0 = -F_1 \sin 30^\circ - F_3 \sin 60^\circ + F_{1,v}\end{aligned}$$

for node 2:

$$\begin{aligned}\sum F_H &= 0 = F_2 + F_1 \cos 30^\circ + F_{2,h} + H_2 \\ \sum F_V &= 0 = F_1 \sin 30^\circ + F_{2,v} + V_2\end{aligned}$$

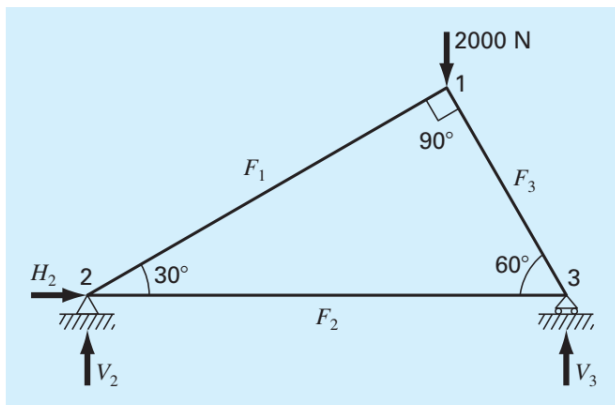


FIGURE P8.10

for node 3:

$$\begin{aligned}\sum F_H &= 0 = -F_2 - F_3 \cos 60^\circ + F_{3,h} \\ \sum F_V &= 0 = F_3 \sin 60^\circ + F_{3,v} + V_3\end{aligned}$$

where $F_{i,h}$ is the external horizontal force applied to node i (where a positive force is from left to right) and $F_{i,v}$ is the external vertical force applied to node i (where a positive force is upward). Thus, in this problem, the 2000-N downward force on node 1 corresponds to $F_{1,v} = -2000$. For this case, all other $F_{i,v}$'s and $F_{i,h}$'s are zero. Express this set of linear algebraic equations in matrix form and then use MATLAB to solve for the unknowns.

8.11 Consider the three mass-four spring system in Fig. P8.11. Determining the equations of motion from $\sum F_x = ma_x$ for each mass using its free-body diagram results in the

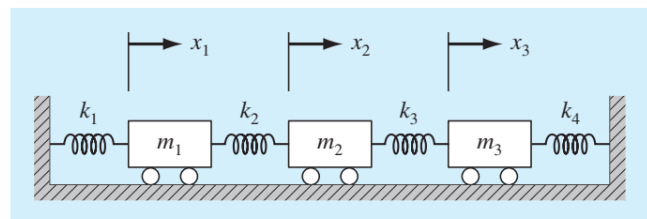


FIGURE P8.11

following differential equations:

$$\ddot{x}_1 + \left(\frac{k_1 + k_2}{m_1} \right) x_1 - \left(\frac{k_2}{m_1} \right) x_2 = 0$$

$$\ddot{x}_2 - \left(\frac{k_2}{m_2} \right) x_1 + \left(\frac{k_2 + k_3}{m_2} \right) x_2 - \left(\frac{k_3}{m_2} \right) x_3 = 0$$

$$\ddot{x}_3 - \left(\frac{k_3}{m_3} \right) x_2 + \left(\frac{k_3 + k_4}{m_3} \right) x_3 = 0$$

where $k_1 = k_4 = 10 \text{ N/m}$, $k_2 = k_3 = 40 \text{ N/m}$, and $m_1 = m_2 = m_3 = 1 \text{ kg}$. The three equations can be written in matrix form:

$$0 = \{\text{Acceleration vector}\} + [k/m \text{ matrix}]\{\text{displacement vector } x\}$$

At a specific time where $x_1 = 0.05 \text{ m}$, $x_2 = 0.04 \text{ m}$, and $x_3 = 0.03 \text{ m}$, this forms a tridiagonal matrix. Use MATLAB to solve for the acceleration of each mass.

8.12 Perform the same computation as in Example 8.2, but use five jumpers with the following characteristics:

Jumper	Mass (kg)	Spring Constant (N/m)	Unstretched Cord Length (m)
1	65	80	10
2	75	40	10
3	60	70	10
4	75	100	10
5	90	20	10

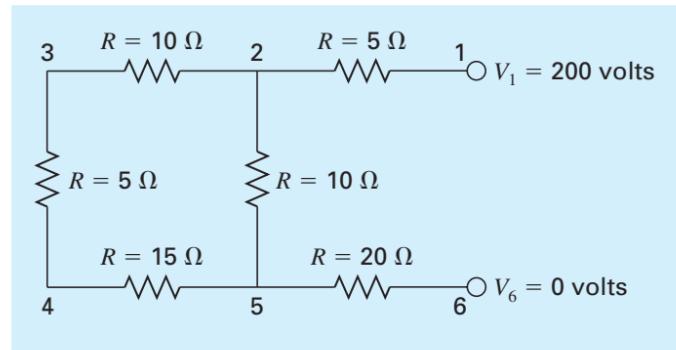


FIGURE P8.14

8.13 Three masses are suspended vertically by a series of identical springs where mass 1 is at the top and mass 3 is at the bottom. If $g = 9.81 \text{ m/s}^2$, $m_1 = 2 \text{ kg}$, $m_2 = 2.5 \text{ kg}$, $m_3 = 3 \text{ kg}$, and the k 's = 15 kg/s^2 , use MATLAB to solve for the displacements x .

8.14 Perform the same computation as in Sec. 8.3, but for the circuit in Fig. P8.14.

8.15 Perform the same computation as in Sec. 8.3, but for the circuit in Fig. P8.15.

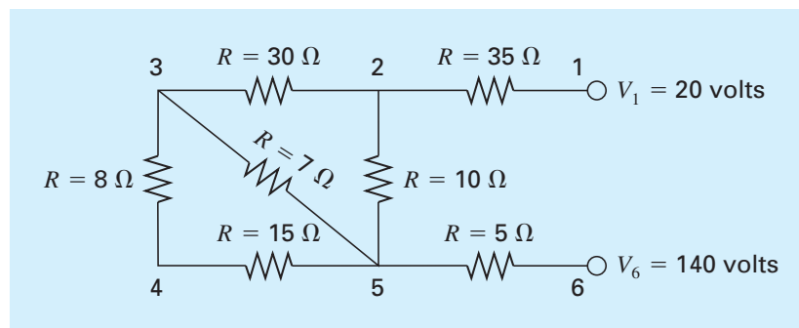


FIGURE P8.15

PROBLEMS

9.1 Determine the number of total flops as a function of the number of equations n for the tridiagonal algorithm (Fig. 9.6).

9.2 Use the graphical method to solve

$$2x_1 - 6x_2 = -18$$

$$-x_1 + 8x_2 = 40$$

Check your results by substituting them back into the equations.

9.3 Given the system of equations

$$0.77x_1 + x_2 = 14.25$$

$$1.2x_1 + 1.7x_2 = 20$$

(a) Solve graphically and check your results by substituting them back into the equations.

(b) On the basis of the graphical solution, what do you expect regarding the condition of the system?

(c) Compute the determinant.

9.4 Given the system of equations

$$2x_2 + 5x_3 = 1$$

$$2x_1 + x_2 + x_3 = 1$$

$$3x_1 + x_2 = 2$$

(a) Compute the determinant.

(b) Use Cramer's rule to solve for the x 's.

(c) Use Gauss elimination with partial pivoting to solve for the x 's. As part of the computation, calculate the determinant in order to verify the value computed in (a)

(d) Substitute your results back into the original equations to check your solution.

9.5 Given the equations

$$0.5x_1 - x_2 = -9.5$$

$$1.02x_1 - 2x_2 = -18.8$$

(a) Solve graphically.

(b) Compute the determinant.

(c) On the basis of (a) and (b), what would you expect regarding the system's condition?

(d) Solve by the elimination of unknowns.

(e) Solve again, but with a_{11} modified slightly to 0.52. Interpret your results.

9.6 Given the equations

$$10x_1 + 2x_2 - x_3 = 27$$

$$-3x_1 - 5x_2 + 2x_3 = -61.5$$

$$x_1 + x_2 + 6x_3 = -21.5$$

(a) Solve by naive Gauss elimination. Show all steps of the computation.

(b) Substitute your results into the original equations to check your answers.

9.7 Given the equations

$$2x_1 - 6x_2 - x_3 = -38$$

$$-3x_1 - x_2 + 7x_3 = -34$$

$$-8x_1 + x_2 - 2x_3 = -20$$

(a) Solve by Gauss elimination with partial pivoting. As part of the computation, use the diagonal elements to calculate the determinant. Show all steps of the computation.

(b) Substitute your results into the original equations to check your answers.

9.8 Perform the same calculations as in Example 9.5, but for the tridiagonal system:

$$\begin{bmatrix} 0.8 & -0.4 & \\ -0.4 & 0.8 & -0.4 \\ & -0.4 & 0.8 \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 41 \\ 25 \\ 105 \end{Bmatrix}$$

9.9 Figure P9.9 shows three reactors linked by pipes. As indicated, the rate of transfer of chemicals through each pipe is equal to a flow rate (Q , with units of cubic meters per second) multiplied by the concentration of the reactor from which the flow originates (c , with units of milligrams per cubic meter). If the system is at a steady state, the transfer into each reactor will balance the transfer out. Develop mass-balance equations for the reactors and solve the three simultaneous linear algebraic equations for their concentrations.

9.10 A civil engineer involved in construction requires 4800, 5800, and 5700 m³ of sand, fine gravel, and coarse gravel, respectively, for a building project. There are three

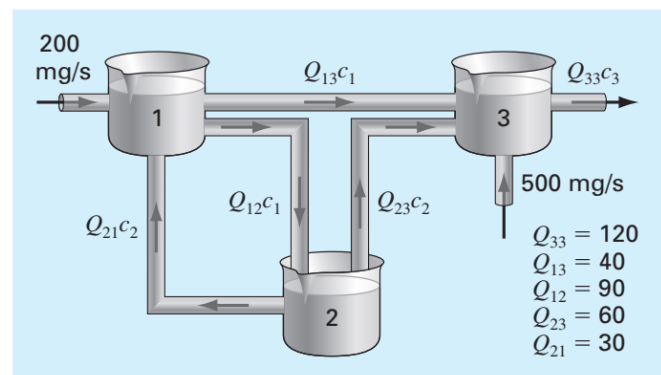


FIGURE P9.9

Three reactors linked by pipes. The rate of mass transfer through each pipe is equal to the product of flow Q and concentration c of the reactor from which the flow originates.

pits from which these materials can be obtained. The composition of these pits is

	Sand %	Fine Gravel %	Coarse Gravel %
Pit1	52	30	18
Pit2	20	50	30
Pit3	25	20	55

How many cubic meters must be hauled from each pit in order to meet the engineer's needs?

9.11 An electrical engineer supervises the production of three types of electrical components. Three kinds of material—metal, plastic, and rubber—are required for production. The amounts needed to produce each component are

Component	Metal (g/ component)	Plastic (g/ component)	Rubber (g/ component)
1	15	0.25	1.0
2	17	0.33	1.2
3	19	0.42	1.6

If totals of 2.12, 0.0434, and 0.164 kg of metal, plastic, and rubber, respectively, are available each day, how many components can be produced per day?

9.12 As described in Sec. 9.4, linear algebraic equations can arise in the solution of differential equations. For example, the following differential equation results from a steady-state mass balance for a chemical in a one-dimensional canal:

$$0 = D \frac{d^2 c}{dx^2} - U \frac{dc}{dx} - kc$$

where c = concentration, t = time, x = distance, D = diffusion coefficient, U = fluid velocity, and k = a first-order decay rate. Convert this differential equation to an equivalent system of simultaneous algebraic equations. Given $D = 2$, $U = 1$, $k = 0.2$, $c(0) = 80$ and $c(10) = 10$, solve these equations from $x = 0$ to 10 and develop a plot of concentration versus distance.

9.13 A stage extraction process is depicted in Fig. P9.13. In such systems, a stream containing a weight fraction y_{in} of a chemical enters from the left at a mass flow rate of F_1 . Simultaneously, a solvent carrying a weight fraction x_{in} of the same chemical enters from the right at a flow rate of F_2 . Thus, for stage i , a mass balance can be represented as

$$F_1 y_{i-1} + F_2 x_{i+1} = F_1 y_i + F_2 x_i \quad (\text{P9.13a})$$

At each stage, an equilibrium is assumed to be established between y_i and x_i as in

$$K = \frac{x_i}{y_i} \quad (\text{P9.13b})$$

where K is called a distribution coefficient. Equation (P9.13b) can be solved for x_i and substituted into Eq. (P9.13a) to yield

$$y_{i-1} - \left(1 + \frac{F_2}{F_1} K\right) y_i + \left(\frac{F_2}{F_1} K\right) y_{i+1} = 0 \quad (\text{P9.13c})$$

If $F_1 = 400$ kg/h, $y_{in} = 0.1$, $F_2 = 800$ kg/h, $x_{in} = 0$, and $K = 5$, determine the values of y_{out} and x_{out} if a five-stage reactor is used. Note that Eq. (P9.13c) must be modified to account for the inflow weight fractions when applied to the first and last stages.

9.14 A peristaltic pump delivers a unit flow (Q_1) of a highly viscous fluid. The network is depicted in Fig. P9.14. Every pipe section has the same length and diameter. The mass and mechanical energy balance can be simplified to obtain the

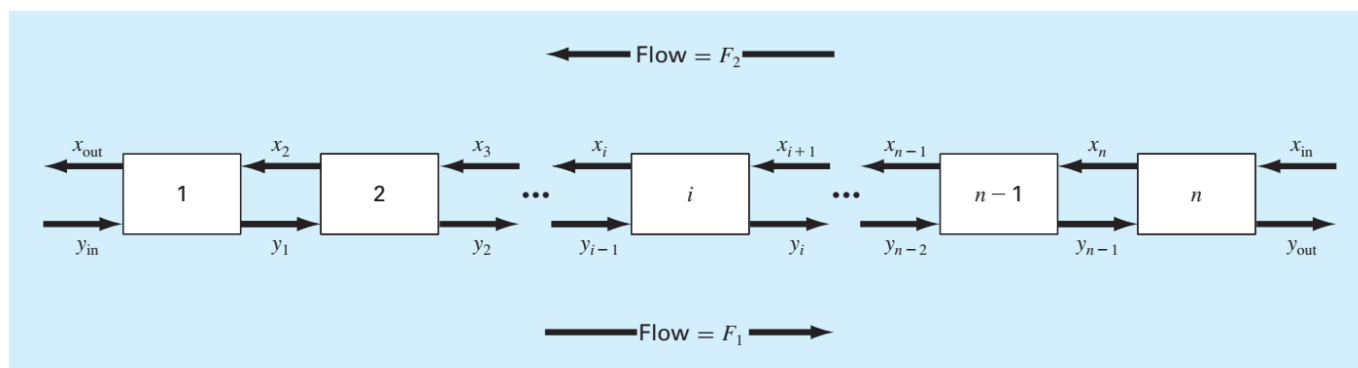


FIGURE P9.13

A stage extraction process.

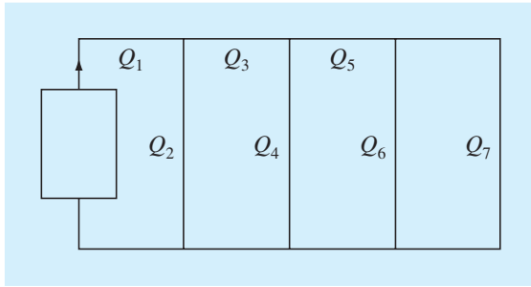


FIGURE P9.14

flows in every pipe. Solve the following system of equations to obtain the flow in every stream.

$$\begin{aligned} Q_3 + 2Q_4 - 2Q_2 &= 0 & Q_1 &= Q_2 + Q_3 \\ Q_5 + 2Q_6 - 2Q_4 &= 0 & Q_3 &= Q_4 + Q_5 \\ 3Q_7 - 2Q_6 &= 0 & Q_5 &= Q_6 + Q_7 \end{aligned}$$

9.15 A truss is loaded as shown in Fig. P9.15. Using the following set of equations, solve for the 10 unknowns, AB , BC , AD , BD , CD , DE , CE , A_x , A_y , and E_y .

$$\begin{aligned} A_x + AD &= 0 & -24 - CD - (4/5)CE &= 0 \\ A_y + AB &= 0 & -AD + DE - (3/5)BD &= 0 \\ 74 + BC + (3/5)BD &= 0 & CD + (4/5)BD &= 0 \\ -AB - (4/5)BD &= 0 & -DE - (3/5)CE &= 0 \\ -BC + (3/5)CE &= 0 & E_y + (4/5)CE &= 0 \end{aligned}$$

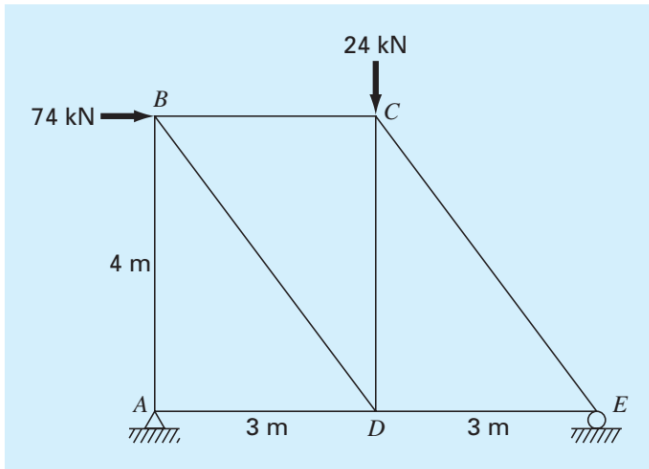


FIGURE P9.15

9.16 A pentadiagonal system with a bandwidth of five can be expressed generally as

$$\begin{bmatrix} f_1 & g_1 & h_1 & & & & \\ e_2 & f_2 & g_2 & h_2 & & & \\ d_3 & e_3 & f_3 & g_3 & h_3 & & \\ & & \cdot & \cdot & \cdot & & \\ & & & \cdot & \cdot & \cdot & \\ & & & & \cdot & \cdot & \cdot \\ & & & & & d_{n-1} & e_{n-1} & f_{n-1} & g_{n-1} \\ & & & & & & d_n & e_n & f_n \end{bmatrix}$$

$$\times \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \cdot \\ \cdot \\ \cdot \\ x_{n-1} \\ x_n \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ \cdot \\ \cdot \\ \cdot \\ r_{n-1} \\ r_n \end{bmatrix}$$

Develop an M-file to efficiently solve such systems without pivoting in a similar fashion to the algorithm used for tridiagonal matrices in Sec. 9.4.1. Test it for the following case:

$$\begin{bmatrix} 8 & -2 & -1 & 0 & 0 \\ -2 & 9 & -4 & -1 & 0 \\ -1 & -3 & 7 & -1 & -2 \\ 0 & -4 & -2 & 12 & -5 \\ 0 & 0 & -7 & -3 & 15 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 1 \\ 1 \\ 5 \end{bmatrix}$$

9.17 Develop an M-file function based on Fig. 9.5 to implement Gauss elimination with partial pivoting. Modify the function so that it computes and returns the determinant (with the correct sign), and detects whether the system is singular based on a near-zero determinant. For the latter, define “near-zero” as being when the absolute value of the determinant is below a tolerance. When this occurs, design the function so that an error message is displayed and the function terminates. Here is the functions first line:

function [x, D] = GaussPivotNew(A, b, tol)

where D = the determinant and tol = the tolerance. Test your program for Prob. 9.5 with tol = 1×10^{-5} .

PROBLEMS

13.1 Repeat Example 13.1 but for three masses with the m 's = 40 kg and the k 's = 240 N/m. Produce a plot like Fig. 13.4 to identify the principle modes of vibration.

13.2 Use the power method to determine the highest eigenvalue and corresponding eigenvector for

$$\begin{bmatrix} 2 - \lambda & 8 & 10 \\ 8 & 4 - \lambda & 5 \\ 10 & 5 & 7 - \lambda \end{bmatrix}$$

13.3 Use the power method to determine the lowest eigenvalue and corresponding eigenvector for the system from Prob. 13.2.

13.4 Derive the set of differential equations for a three mass–four spring system (Fig. P13.4) that describes their time motion. Write the three differential equations in matrix form

$$\{\text{Acceleration vector}\} + [k/m \text{ matrix}] \{\text{displacement vector } x\} = 0$$

Note each equation has been divided by the mass. Solve for the eigenvalues and natural frequencies for the following values of mass and spring constants: $k_1 = k_4 = 15$ N/m, $k_2 = k_3 = 35$ N/m, and $m_1 = m_2 = m_3 = 1.5$ kg.

13.5 Consider the mass-spring system in Fig. P13.5. The frequencies for the mass vibrations can be determined by solving for the eigenvalues and by applying $M\ddot{x} + kx = 0$, which yields

$$\begin{bmatrix} m_1 & 0 & 0 \\ 0 & m_2 & 0 \\ 0 & 0 & m_3 \end{bmatrix} \begin{Bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \\ \ddot{x}_3 \end{Bmatrix} + \begin{bmatrix} 2k & -k & -k \\ -k & 2k & -k \\ -k & -k & 2k \end{bmatrix} \begin{Bmatrix} x_1 \\ x_2 \\ x_3 \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$$

Applying the guess $x = x_0 e^{i\omega t}$ as a solution, we get the following matrix:

$$\begin{bmatrix} 2k - m_1\omega^2 & -k & -k \\ -k & 2k - m_2\omega^2 & -k \\ -k & -k & 2k - m_3\omega^2 \end{bmatrix} \begin{Bmatrix} x_{01} \\ x_{02} \\ x_{03} \end{Bmatrix} e^{i\omega t} = \begin{Bmatrix} 0 \\ 0 \\ 0 \end{Bmatrix}$$

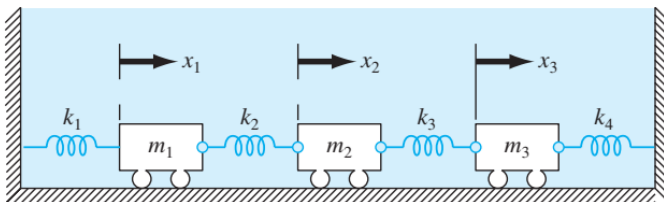


FIGURE P13.4

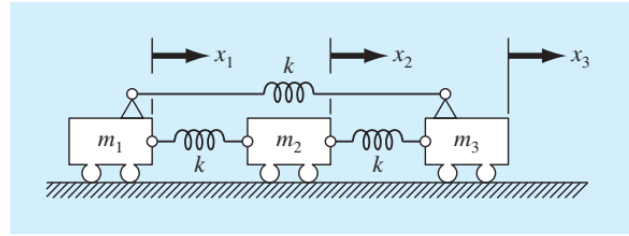


FIGURE P13.5

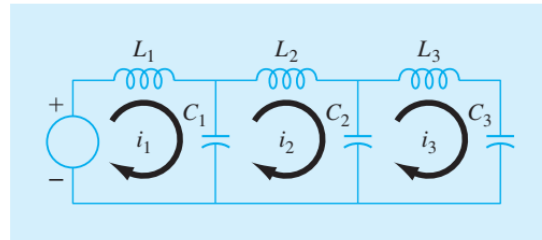


FIGURE P13.6

Use MATLAB's `eig` command to solve for the eigenvalues of the $k - m\omega^2$ matrix above. Then use these eigenvalues to solve for the frequencies (ω). Let $m_1 = m_2 = m_3 = 1$ kg, and $k = 2$ N/m.

13.6 As displayed in Fig. P13.6, an LC circuit can be modeled by the following system of differential equations:

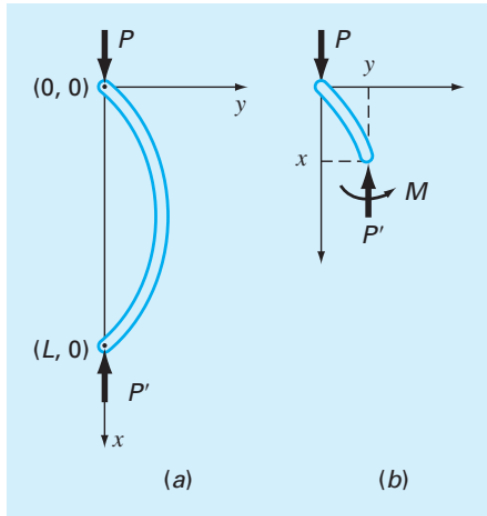
$$\begin{aligned} L_1 \frac{d^2 i_1}{dt^2} + \frac{1}{C_1} (i_1 - i_2) &= 0 \\ L_2 \frac{d^2 i_2}{dt^2} + \frac{1}{C_2} (i_2 - i_3) - \frac{1}{C_1} (i_1 - i_2) &= 0 \\ L_3 \frac{d^2 i_3}{dt^2} + \frac{1}{C_3} i_3 - \frac{1}{C_2} (i_2 - i_3) &= 0 \end{aligned}$$

where L = inductance (H), t = time (s), i = current (A), and C = capacitance (F). Assuming that a solution is of the form $i_j = I_j \sin(\omega t)$, determine the eigenvalues and eigenvectors for this system with $L = 1$ H and $C = 0.25$ F. Draw the network, illustrating how the currents oscillate in their primary modes.

13.7 Repeat Prob. 13.6 but with only two loops. That is, omit the i_3 loop. Draw the network, illustrating how the currents oscillate in their primary modes.

13.8 Repeat the problem in Sec. 13.5 but leave off the third floor.

13.9 Repeat the problem in Sec. 13.5 but add a fourth floor with $m_4 = 6,000$ and $k_4 = 1,200$ kN/m.

**FIGURE P13.10**

(a) A slender rod. (b) A freebody diagram of a rod.

13.10 The curvature of a slender column subject to an axial load P (Fig. P13.10) can be modeled by

$$\frac{d^2 y}{dx^2} + p^2 y = 0$$

where

$$p^2 = \frac{P}{EI}$$

where E = the modulus of elasticity, and I = the moment of inertia of the cross section about its neutral axis.

This model can be converted into an eigenvalue problem by substituting a centered finite-difference approximation for the second derivative to give

$$\frac{y_{i+1} - 2y_i + y_{i-1}}{\Delta x^2} + p^2 y_i = 0$$

where i = a node located at a position along the rod's interior, and Δx = the spacing between nodes. This equation can be expressed as

$$y_{i-1} - (2 - \Delta x^2 p^2) y_i + y_{i+1} = 0$$

Writing this equation for a series of interior nodes along the axis of the column yields a homogeneous system of equations. For example, if the column is divided into five segments (i.e., four interior nodes), the result is

$$\begin{bmatrix} (2 - \Delta x^2 p^2) & -1 & 0 & 0 \\ -1 & (2 - \Delta x^2 p^2) & -1 & 0 \\ 0 & -1 & (2 - \Delta x^2 p^2) & -1 \\ 0 & 0 & -1 & (2 - \Delta x^2 p^2) \end{bmatrix} \begin{Bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{Bmatrix} = 0$$

An axially loaded wooden column has the following characteristics: $E = 10 \times 10^9$ Pa, $I = 1.25 \times 10^{-5}$ m⁴, and $L = 3$ m. For the five-segment, four-node representation:

- Implement the polynomial method with MATLAB to determine the eigenvalues for this system.
- Use the MATLAB `eig` function to determine the eigenvalues and eigenvectors.
- Use the power method to determine the largest eigenvalue and its corresponding eigenvector.

13.11 A system of two homogeneous linear ordinary differential equations with constant coefficients can be written as

$$\begin{aligned} \frac{dy_1}{dt} &= -5y_1 + 3y_2, & y_1(0) &= 50 \\ \frac{dy_2}{dt} &= 100y_1 - 301y_2, & y_2(0) &= 100 \end{aligned}$$

If you have taken a course in differential equations, you know that the solutions for such equations have the form

$$y_i = ce^{\lambda t}$$

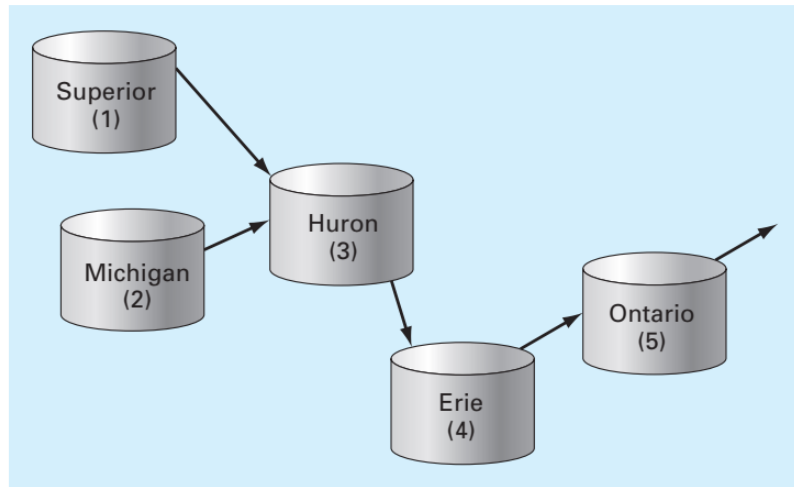
where c and λ are constants to be determined. Substituting this solution and its derivative into the original equations converts the system into an eigenvalue problem. The resulting eigenvalues and eigenvectors can then be used to derive the general solution to the differential equations. For example, for the two-equation case, the general solution can be written in terms of vectors as

$$\{y\} = c_1 \{v_1\} e^{\lambda_1 t} + c_2 \{v_2\} e^{\lambda_2 t}$$

where $\{v_i\}$ = the eigenvector corresponding to the i^{th} eigenvalue (λ_i) and the c 's are unknown coefficients that can be determined with the initial conditions.

- Convert the system into an eigenvalue problem.
- Use MATLAB to solve for the eigenvalues and eigenvectors.
- Employ the results of (b) and the initial conditions to determine the general solution.
- Develop a MATLAB plot of the solution for $t = 0$ to 1.

13.12 Water flows between the North American Great Lakes as depicted in Fig. P13.12. Based on mass balances, the following differential equations can be written for the concentrations in each of the lakes for a pollutant that decays with first-order kinetics:

**FIGURE P13.12**

The North American Great Lakes. The arrows indicate how water flows between the lakes.

$$\frac{dc_1}{dt} = -(0.0056 + k)c_1$$

$$\frac{dc_2}{dt} = -(0.01 + k)c_2$$

$$\frac{dc_3}{dt} = 0.01902c_1 + 0.01387c_2 - (0.047 + k)c_3$$

$$\frac{dc_4}{dt} = 0.33597c_3 - (0.376 + k)c_4$$

$$\frac{dc_5}{dt} = 0.11364c_4 - (0.133 + k)c_5$$

where k = the first-order decay rate (/yr), which is equal to $0.69315/(\text{half-life})$. Note that the constants in each of the

equations account for the flow between the lakes. Due to the testing of nuclear weapons in the atmosphere, the concentrations of strontium-90 (^{90}Sr) in the five lakes in 1963 were approximately $\{c\} = \{17.7 \ 30.5 \ 43.9 \ 136.3 \ 30.1\}^T$ in units of Bq/m^3 . Assuming that no additional ^{90}Sr entered the system thereafter, use MATLAB and the approach outlined in Prob. 13.11 to compute the concentrations in each of the lakes from 1963 through 2010. Note that ^{90}Sr has a half-life of 28.8 years.

13.13 Develop an M-file function to determine the largest eigenvalue and its associated eigenvector with the power method. Test the program by duplicating Example 13.3 and then use it to solve Prob. 13.2.

PROBLEMS

14.1 Given the data

0.90	1.42	1.30	1.55	1.63
1.32	1.35	1.47	1.95	1.66
1.96	1.47	1.92	1.35	1.05
1.85	1.74	1.65	1.78	1.71
2.29	1.82	2.06	2.14	1.27

Determine **(a)** the mean, **(b)** median, **(c)** mode, **(d)** range, **(e)** standard deviation, **(f)** variance, and **(g)** coefficient of variation.

14.2 Construct a histogram from the data from Prob. 14.1. Use a range from 0.8 to 2.4 with intervals of 0.2.

14.3 Given the data

29.65	28.55	28.65	30.15	29.35	29.75	29.25
30.65	28.15	29.85	29.05	30.25	30.85	28.75
29.65	30.45	29.15	30.45	33.65	29.35	29.75
31.25	29.45	30.15	29.65	30.55	29.65	29.25

Determine **(a)** the mean, **(b)** median, **(c)** mode, **(d)** range, **(e)** standard deviation, **(f)** variance, and **(g)** coefficient of variation.

(h) Construct a histogram. Use a range from 28 to 34 with increments of 0.4.

(i) Assuming that the distribution is normal, and that your estimate of the standard deviation is valid, compute the range (i.e., the lower and the upper values) that encompasses 68% of the readings. Determine whether this is a valid estimate for the data in this problem.

14.4 Using the same approach as was employed to derive Eqs. (14.15) and (14.16), derive the least-squares fit of the following model:

$$y = a_1x + e$$

That is, determine the slope that results in the least-squares fit for a straight line with a zero intercept. Fit the following data with this model and display the result graphically.

x	2	4	6	7	10	11	14	17	20
y	4	5	6	5	8	8	6	9	12

14.5 Use least-squares regression to fit a straight line to

x	0	2	4	6	9	11	12	15	17	19
y	5	6	7	6	9	8	8	10	12	12

Along with the slope and intercept, compute the standard error of the estimate and the correlation coefficient. Plot the data and the regression line. Then repeat the problem, but regress x versus y —that is, switch the variables. Interpret your results.

14.6 Fit a power model to the data from Table 14.1, but use natural logarithms to perform the transformations.

14.7 The following data were gathered to determine the relationship between pressure and temperature of a fixed volume of 1 kg of nitrogen. The volume is 10 m^3 .

T, °C	−40	0	40	80	120	160
p, N/m²	6900	8100	9350	10,500	11,700	12,800

Employ the ideal gas law $pV = nRT$ to determine R on the basis of these data. Note that for the law, T must be expressed in kelvins.

14.8 Beyond the examples in Fig. 14.13, there are other models that can be linearized using transformations. For example,

$$y = \alpha_4 x e^{\beta_4 x}$$

Linearize this model and use it to estimate α_4 and β_4 based on the following data. Develop a plot of your fit along with the data.

x	0.1	0.2	0.4	0.6	0.9	1.3	1.5	1.7	1.8
y	0.75	1.25	1.45	1.25	0.85	0.55	0.35	0.28	0.18

14.9 The concentration of *E. coli* bacteria in a swimming area is monitored after a storm:

t (hr)	4	8	12	16	20	24
c (CFU/100 mL)	1600	1320	1000	890	650	560

The time is measured in hours following the end of the storm and the unit CFU is a “colony forming unit.” Use this data to estimate (a) the concentration at the end of the storm ($t = 0$) and (b) the time at which the concentration will reach 200 CFU/100 mL. Note that your choice of model should be consistent with the fact that negative concentrations are impossible and that the bacteria concentration always decreases with time.

14.10 Rather than using the base- e exponential model (Eq. 14.22), a common alternative is to employ a base-10 model:

$$y = \alpha_5 10^{\beta_5 x}$$

When used for curve fitting, this equation yields identical results to the base- e version, but the value of the exponent parameter (β_5) will differ from that estimated with Eq. 14.22 (β_1). Use the base-10 version to solve Prob. 14.9. In addition, develop a formulation to relate β_1 to β_5 .

14.11 Determine an equation to predict metabolism rate as a function of mass based on the following data. Use it to predict the metabolism rate of a 200-kg tiger.

Animal	Mass (kg)	Metabolism (watts)
Cow	400	270
Human	70	82
Sheep	45	50
Hen	2	4.8
Rat	0.3	1.45
Dove	0.16	0.97

14.12 On average, the surface area A of human beings is related to weight W and height H . Measurements on a number of individuals of height 180 cm and different weights (kg) give values of A (m^2) in the following table:

W (kg)	70	75	77	80	82	84	87	90
A (m^2)	2.10	2.12	2.15	2.20	2.22	2.23	2.26	2.30

Show that a power law $A = aW^b$ fits these data reasonably well. Evaluate the constants a and b , and predict what the surface area is for a 95-kg person.

14.13 Fit an exponential model to

x	0.4	0.8	1.2	1.6	2	2.3
y	800	985	1490	1950	2850	3600

Plot the data and the equation on both standard and semi-logarithmic graphs with the MATLAB `subplot` function.

14.14 An investigator has reported the data tabulated below for an experiment to determine the growth rate of bacteria k (per d) as a function of oxygen concentration c (mg/L). It is known that such data can be modeled by the following equation:

$$k = \frac{k_{\max} c^2}{c_s + c^2}$$

where c_s and $k_{\max}$ are parameters. Use a transformation to linearize this equation. Then use linear regression to estimate c_s and $k_{\max}$ and predict the growth rate at $c = 2$ mg/L.

c	0.5	0.8	1.5	2.5	4
k	1.1	2.5	5.3	7.6	8.9

14.15 Develop an M-file function to compute descriptive statistics for a vector of values. Have the function determine and display number of values, mean, median, mode, range, standard deviation, variance, and coefficient of variation. In addition, have it generate a histogram. Test it with the data from Prob. 14.3.

14.16 Modify the `linregr` function in Fig. 14.15 so that it (a) computes and returns the standard error of the estimate, and (b) uses the `subplot` function to also display a plot of the residuals (the predicted minus the measured y) versus x .

14.17 Develop an M-file function to fit a power model. Have the function return the best-fit coefficient α_2 and power β_2 along with the r^2 for the untransformed model. In addition, use the `subplot` function to display graphs of both the transformed and untransformed equations along with the data. Test it with the data from Prob. 14.11.

14.18 The following data show the relationship between the viscosity of SAE 70 oil and temperature. After taking the log of the data, use linear regression to find the equation of the line that best fits the data and the r^2 value.

Temperature, °C	26.67	93.33	148.89	315.56
Viscosity, μ , N·s/m ²	1.35	0.085	0.012	0.00075

14.19 You perform experiments and determine the following values of heat capacity c at various temperatures T for a gas:

T	−50	−30	0	60	90	110
c	1250	1280	1350	1480	1580	1700

Use regression to determine a model to predict c as a function of T .

14.20 It is known that the tensile strength of a plastic increases as a function of the time it is heat treated. The following data are collected:

Time	10	15	20	25	40	50	55	60	75
Tensile Strength	5	20	18	40	33	54	70	60	78

- (a) Fit a straight line to these data and use the equation to determine the tensile strength at a time of 32 min.
- (b) Repeat the analysis but for a straight line with a zero intercept.

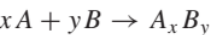
14.21 The following data were taken from a stirred tank reactor for the reaction $A \rightarrow B$. Use the data to determine the best possible estimates for k_{01} and E_1 for the following kinetic model:

$$-\frac{dA}{dt} = k_{01}e^{-E_1/RT}A$$

where R is the gas constant and equals 0.00198 kcal/mol/K.

$-dA/dt$ (moles/L/s)	460	960	2485	1600	1245
A (moles/L)	200	150	50	20	10
T (K)	280	320	450	500	550

14.22 Concentration data were collected at 15 time points for the polymerization reaction:



We assume the reaction occurs via a complex mechanism consisting of many steps. Several models have been hypothesized, and the sum of the squares of the residuals had been calculated for the fits of the models of the data. The results

are shown below. Which model best describes the data (statistically)? Explain your choice.

	Model A	Model B	Model C
S_r	135	105	100
Number of Model Parameters Fit	2	3	5

14.23 Below are data taken from a batch reactor of bacterial growth (after lag phase was over). The bacteria are allowed to grow as fast as possible for the first 2.5 hours, and then they are induced to produce a recombinant protein, the production of which slows the bacterial growth significantly. The theoretical growth of bacteria can be described by

$$\frac{dX}{dt} = \mu X$$

where X is the number of bacteria, and μ is the specific growth rate of the bacteria during exponential growth. Based on the data, estimate the specific growth rate of the bacteria during the first 2 hours of growth and during the next 4 hours of growth.

Time, h	0	1	2	3	4	5	6
[Cells], g/L	0.100	0.335	1.102	1.655	2.453	3.702	5.460

14.24 A transportation engineering study was conducted to determine the proper design of bike lanes. Data were gathered on bike-lane widths and average distance between bikes and passing cars. The data from 9 streets are

Distance, m	2.4	1.5	2.4	1.8	1.8	2.9	1.2	3	1.2
Lane Width, m	2.9	2.1	2.3	2.1	1.8	2.7	1.5	2.9	1.5

- (a) Plot the data.
- (b) Fit a straight line to the data with linear regression. Add this line to the plot.
- (c) If the minimum safe average distance between bikes and passing cars is considered to be 1.8 m, determine the corresponding minimum lane width.

14.25 In water-resources engineering, the sizing of reservoirs depends on accurate estimates of water flow in the river that is being impounded. For some rivers, long-term historical records of such flow data are difficult to obtain. In contrast, meteorological data on precipitation are often available for many years past. Therefore, it is often useful to

determine a relationship between flow and precipitation. This relationship can then be used to estimate flows for years when only precipitation measurements were made. The following data are available for a river that is to be dammed:

Precip., cm/yr	88.9	108.5	104.1	139.7	127	94	116.8	99.1
Flow, m³/s	14.6	16.7	15.3	23.2	19.5	16.1	18.1	16.6

- Plot the data.
- Fit a straight line to the data with linear regression. Superimpose this line on your plot.
- Use the best-fit line to predict the annual water flow if the precipitation is 120 cm.
- If the drainage area is 1100 km², estimate what fraction of the precipitation is lost via processes such as evaporation, deep groundwater infiltration, and consumptive use.

14.26 The mast of a sailboat has a cross-sectional area of 10.65 cm² and is constructed of an experimental aluminum alloy. Tests were performed to define the relationship between stress and strain. The test results are

Strain, cm/cm	0.0032	0.0045	0.0055	0.0016	0.0085	0.0005
Stress, N/cm²	4970	5170	5500	3590	6900	1240

The stress caused by wind can be computed as F/A_c where F = force in the mast and A_c = mast's cross-sectional area. This value can then be substituted into Hooke's law to determine the mast's deflection, $\Delta L = \text{strain} \times L$, where L = the mast's length. If the wind force is 25,000 N, use the data to estimate the deflection of a 9-m mast.

14.27 The following data were taken from an experiment that measured the current in a wire for various imposed voltages:

V, V	2	3	4	5	7	10
i, A	5.2	7.8	10.7	13	19.3	27.5

- On the basis of a linear regression of this data, determine current for a voltage of 3.5 V. Plot the line and the data and evaluate the fit.
- Redo the regression and force the intercept to be zero.

14.28 An experiment is performed to determine the % elongation of electrical conducting material as a function of temperature. The resulting data are listed below. Predict the % elongation for a temperature of 400 °C.

Temperature, °C	200	250	300	375	425	475	600
% Elongation	7.5	8.6	8.7	10	11.3	12.7	15.3

14.29 The population p of a small community on the outskirts of a city grows rapidly over a 20-year period:

t	0	5	10	15	20
p	100	200	450	950	2000

As an engineer working for a utility company, you must forecast the population 5 years into the future in order to anticipate the demand for power. Employ an exponential model and linear regression to make this prediction.

14.30 The velocity u of air flowing past a flat surface is measured at several distances y away from the surface. Fit a curve to this data assuming that the velocity is zero at the surface ($y = 0$). Use your result to determine the shear stress (du/dy) at the surface.

y, m	0.002	0.006	0.012	0.018	0.024
u, m/s	0.287	0.899	1.915	3.048	4.299

14.31 Andrade's equation has been proposed as a model of the effect of temperature on viscosity:

$$\mu = De^{B/T_a}$$

where μ = dynamic viscosity of water (10^{-3} N·s/m²), T_a = absolute temperature (K), and D and B are parameters. Fit this model to the following data for water:

T	0	5	10	20	30	40
μ	1.787	1.519	1.307	1.002	0.7975	0.6529

14.32 Perform the same computation as in Example 14.2, but in addition to the drag coefficient, also vary the mass uniformly by $\pm 10\%$.

14.33 Perform the same computation as in Example 14.3, but in addition to the drag coefficient, also vary the mass normally around its mean value with a coefficient of variation of 5.7887%.

14.34 Manning's formula for a rectangular channel can be written as

$$Q = \frac{1}{n_m} \frac{(BH)^{5/3}}{(B + 2H)^{2/3}} \sqrt{S}$$

where Q = flow (m^3/s), n_m = a roughness coefficient, B = width (m), H = depth (m), and S = slope. You are applying this formula to a stream where you know that the width = 20 m and the depth = 0.3 m. Unfortunately, you know the roughness and the slope to only a $\pm 10\%$ precision. That is, you know that the roughness is about 0.03 with a range from 0.027 to 0.033 and the slope is 0.0003 with a range from 0.00027 to 0.00033. Assuming uniform distributions, use a Monte Carlo analysis with $n = 10,000$ to estimate the distribution of flow.

14.35 A Monte Carlo analysis can be used for optimization. For example, the trajectory of a ball can be computed with

$$y = (\tan\theta_0)x - \frac{g}{2v_0^2 \cos^2\theta_0}x^2 + y_0 \quad (\text{P14.35})$$

where y = the height (m), θ_0 = the initial angle (radians), v_0 = the initial velocity (m/s), g = the gravitational constant = 9.81 m/s^2 , and y_0 = the initial height (m). Given $y_0 = 1 \text{ m}$, $v_0 = 25 \text{ m/s}$, and $\theta_0 = 50^\circ$, determine the maximum height and the corresponding x distance (**a**) analytically with calculus and (**b**) numerically with Monte Carlo simulation. For the latter, develop a script that generates a vector of 10,000 uniformly-distributed values of x between 0 and 60 m. Use this vector and Eq. P14.35 to generate a vector of heights. Then, employ the `max` function to determine the maximum height and the associated x distance.

PROBLEMS

15.1 Fit a parabola to the data from Table 14.1. Determine the r^2 for the fit and comment on the efficacy of the result.

15.2 Using the same approach as was employed to derive Eqs. (14.15) and (14.16), derive the least-squares fit of the following model:

$$y = a_1x + a_2x^2 + e$$

That is, determine the coefficients that result in the least-squares fit for a second-order polynomial with a zero intercept. Test the approach by using it to fit the data from Table 14.1.

15.3 Fit a cubic polynomial to the following data:

x	3	4	5	7	8	9	11	12
y	1.6	3.6	4.4	3.4	2.2	2.8	3.8	4.6

Along with the coefficients, determine r^2 and $s_{y/x}$.

15.4 Develop an M-file to implement polynomial regression. Pass the M-file two vectors holding the x and y values along with the desired order m . Test it by solving Prob. 15.3.

15.5 For the data from Table P15.5, use polynomial regression to derive a predictive equation for dissolved oxygen concentration as a function of temperature for the case where the chloride concentration is equal to zero. Employ a polynomial that is of sufficiently high order that the predictions match the number of significant digits displayed in the table.

15.6 Use multiple linear regression to derive a predictive equation for dissolved oxygen concentration as a function of temperature and chloride based on the data from Table P15.5. Use the equation to estimate the concentration of dissolved oxygen for a chloride concentration of 15 g/L at $T = 12^\circ\text{C}$. Note that the true value is 9.09 mg/L . Compute the percent

TABLE P15.5 Dissolved oxygen concentration in water as a function of temperature (°C) and chloride concentration (g/L).

$T, ^\circ\text{C}$	Dissolved Oxygen (mg/L) for Temperature (°C) and Concentration of Chloride (g/L)		
	$c = 0 \text{ g/L}$	$c = 10 \text{ g/L}$	$c = 20 \text{ g/L}$
0	14.6	12.9	11.4
5	12.8	11.3	10.3
10	11.3	10.1	8.96
15	10.1	9.03	8.08
20	9.09	8.17	7.35
25	8.26	7.46	6.73
30	7.56	6.85	6.20

relative error for your prediction. Explain possible causes for the discrepancy.

15.7 As compared with the models from Probs. 15.5 and 15.6, a somewhat more sophisticated model that accounts for the effect of both temperature and chloride on dissolved oxygen saturation can be hypothesized as being of the form

$$o = f_3(T) + f_1(c)$$

That is, a third-order polynomial in temperature and a linear relationship in chloride is assumed to yield superior results. Use the general linear least-squares approach to fit this model to the data in Table P15.5. Use the resulting equation to estimate the dissolved oxygen concentration for a chloride concentration of 15 g/L at $T = 12^\circ\text{C}$. Note that the true value is 9.09 mg/L. Compute the percent relative error for your prediction.

15.8 Use multiple linear regression to fit

x_1	0	1	1	2	2	3	3	4	4
x_2	0	1	2	1	2	1	2	1	2
y	15.1	17.9	12.7	25.6	20.5	35.1	29.7	45.4	40.2

Compute the coefficients, the standard error of the estimate, and the correlation coefficient.

15.9 The following data were collected for the steady flow of water in a concrete circular pipe:

Experiment	Diameter, m	Slope, m/m	Flow, m^3/s
1	0.3	0.001	0.04
2	0.6	0.001	0.24
3	0.9	0.001	0.69
4	0.3	0.01	0.13
5	0.6	0.01	0.82
6	0.9	0.01	2.38
7	0.3	0.05	0.31
8	0.6	0.05	1.95
9	0.9	0.05	5.66

Use multiple linear regression to fit the following model to this data:

$$Q = \alpha_0 D^{\alpha_1} S^{\alpha_2}$$

where Q = flow, D = diameter, and S = slope.

15.10 Three disease-carrying organisms decay exponentially in seawater according to the following model:

$$p(t) = Ae^{-1.5t} + Be^{-0.3t} + Ce^{-0.05t}$$

Estimate the initial concentration of each organism (A , B , and C) given the following measurements:

t	0.5	1	2	3	4	5	6	7	9
$p(t)$	6	4.4	3.2	2.7	2	1.9	1.7	1.4	1.1

15.11 The following model is used to represent the effect of solar radiation on the photosynthesis rate of aquatic plants:

$$P = P_m \frac{I}{I_{sat}} e^{-\frac{I}{I_{sat}} + 1}$$

where P = the photosynthesis rate ($\text{mg m}^{-3}\text{d}^{-1}$), P_m = the maximum photosynthesis rate ($\text{mg m}^{-3}\text{d}^{-1}$), I = solar radiation ($\mu\text{E m}^{-2}\text{s}^{-1}$), and I_{sat} = optimal solar radiation ($\mu\text{E m}^{-2}\text{s}^{-1}$). Use nonlinear regression to evaluate P_m and I_{sat} based on the following data:

I	50	80	130	200	250	350	450	550	700
P	99	177	202	248	229	219	173	142	72

15.12 The following data are provided

x	1	2	3	4	5
y	2.2	2.8	3.6	4.5	5.5

Fit the following model to this data using MATLAB and the general linear least-squares model

$$y = a + bx + \frac{c}{x}$$

15.13 In Prob. 14.8 we used transformations to linearize and fit the following model:

$$y = \alpha_4 x e^{\beta_4 x}$$

Use nonlinear regression to estimate α_4 and β_4 based on the following data. Develop a plot of your fit along with the data.

x	0.1	0.2	0.4	0.6	0.9	1.3	1.5	1.7	1.8
y	0.75	1.25	1.45	1.25	0.85	0.55	0.35	0.28	0.18

15.14 Enzymatic reactions are used extensively to characterize biologically mediated reactions. The following is an example of a model that is used to fit such reactions:

$$v_0 = \frac{k_m [S]^3}{K + [S]^3}$$

where v_0 = the initial rate of the reaction (M/s), $[S]$ = the substrate concentration (M), and k_m and K are parameters. The following data can be fit with this model:

$[S], \text{M}$	$v_0, \text{M/s}$
0.01	6.078×10^{-11}
0.05	7.595×10^{-9}
0.1	6.063×10^{-8}
0.5	5.788×10^{-6}
1	1.737×10^{-5}
5	2.423×10^{-5}
10	2.430×10^{-5}
50	2.431×10^{-5}
100	2.431×10^{-5}

- (a) Use a transformation to linearize the model and evaluate the parameters. Display the data and the model fit on a graph.
 (b) Perform the same evaluation as in (a) but use nonlinear regression.

15.15 Given the data

x	5	10	15	20	25	30	35	40	45	50
y	17	24	31	33	37	37	40	40	42	41

use least-squares regression to fit (a) a straight line, (b) a power equation, (c) a saturation-growth-rate equation, and (d) a parabola. For (b) and (c), employ transformations to linearize the data. Plot the data along with all the curves. Is any one of the curves superior? If so, justify.

15.16 The following data represent the bacterial growth in a liquid culture over of number of days:

Day	0	4	8	12	16	20
Amount $\times 10^6$	67.38	74.67	82.74	91.69	101.60	112.58

Find a best-fit equation to the data trend. Try several possibilities—linear, quadratic, and exponential. Determine the best equation to predict the amount of bacteria after 30 days.

15.17 Dynamic viscosity of water $\mu (10^{-3} \text{ N} \cdot \text{s/m}^2)$ is related to temperature $T (^\circ\text{C})$ in the following manner:

T	0	5	10	20	30	40
μ	1.787	1.519	1.307	1.002	0.7975	0.6529

- (a) Plot this data.
 (b) Use linear interpolation to predict μ at $T = 7.5^\circ\text{C}$.
 (c) Use polynomial regression to fit a parabola to the data in order to make the same prediction.

15.18 Use the following set of pressure-volume data to find the best possible virial constants (A_1 and A_2) for the following equation of state. $R = 82.05 \text{ mL atm/gmol K}$, and $T = 303 \text{ K}$.

$$\frac{PV}{RT} = 1 + \frac{A_1}{V} + \frac{A_2}{V^2}$$

$P (\text{atm})$	0.985	1.108	1.363	1.631
$V (\text{mL})$	25,000	22,200	18,000	15,000

15.19 Environmental scientists and engineers dealing with the impacts of acid rain must determine the value of the ion product of water K_w as a function of temperature. Scientists have suggested the following equation to model this relationship:

$$-\log_{10} K_w = \frac{a}{T_a} + b \log_{10} T_a + c T_a + d$$

where T_a = absolute temperature (K), and a , b , c , and d are parameters. Employ the following data and regression to

estimate the parameters with MATLAB. Also, generate a plot of predicted K_w versus the data.

T ($^{\circ}\text{C}$)	K_w
0	1.164×10^{-15}
10	2.950×10^{-15}
20	6.846×10^{-15}
30	1.467×10^{-14}
40	2.929×10^{-14}

15.20 The distance required to stop an automobile consists of both thinking and braking components, each of which is a function of its speed. The following experimental data were collected to quantify this relationship. Develop best-fit equations for both the thinking and braking components. Use these equations to estimate the total stopping distance for a car traveling at 110 km/h.

Speed, km/h	30	45	60	75	90	120
Thinking, m	5.6	8.5	11.1	14.5	16.7	22.4
Braking, m	5.0	12.3	21.0	32.9	47.6	84.7

15.21 An investigator has reported the data tabulated below. It is known that such data can be modeled by the following equation

$$x = e^{(y-b)/a}$$

where a and b are parameters. Use nonlinear regression to determine a and b . Based on your analysis predict y at $x = 2.6$.

x	1	2	3	4	5
y	0.5	2	2.9	3.5	4

15.22 It is known that the data tabulated below can be modeled by the following equation

$$y = \left(\frac{a + \sqrt{x}}{b\sqrt{x}} \right)^2$$

Use nonlinear regression to determine the parameters a and b . Based on your analysis predict y at $x = 1.6$.

x	0.5	1	2	3	4
y	10.4	5.8	3.3	2.4	2

15.23 An investigator has reported the data tabulated below for an experiment to determine the growth rate of bacteria k (per d), as a function of oxygen concentration c (mg/L). It is known that such data can be modeled by the following equation:

$$k = \frac{k_{\max} c^2}{c_s + c^2}$$

Use nonlinear regression to estimate c_s and $k_{\max}$ and predict the growth rate at $c = 2$ mg/L.

c	0.5	0.8	1.5	2.5	4
k	1.1	2.4	5.3	7.6	8.9

15.24 A material is tested for cyclic fatigue failure whereby a stress, in MPa, is applied to the material and the number of cycles needed to cause failure is measured. The results are in the table below. Use nonlinear regression to fit a power model to this data.

N , cycles	1	10	100	1000	10,000	100,000	1,000,000
Stress, MPa	1100	1000	925	800	625	550	420

15.25 The following data shows the relationship between the viscosity of SAE 70 oil and temperature. Use nonlinear regression to fit a power equation to this data.

Temperature, T , $^{\circ}\text{C}$	26.67	93.33	148.89	315.56
Viscosity, μ , $\text{N}\cdot\text{s}/\text{m}^2$	1.35	0.085	0.012	0.00075

15.26 The concentration of *E. coli* bacteria in a swimming area is monitored after a storm:

t (hr)	4	8	12	16	20	24
c (CFU/100 mL)	1590	1320	1000	900	650	560

The time is measured in hours following the end of the storm and the unit CFU is a "colony forming unit." Employ nonlinear regression to fit an exponential model (Eq. 14.22) to this data. Use the model to estimate (a) the concentration at the end of the storm ($t = 0$) and (b) the time at which the concentration will reach 200 CFU/100 mL.

15.27 Use the following set of pressure-volume data to find the best possible virial constants (A_1 and A_2) for the equation of state shown below. $R = 82.05$ mL atm/gmol K and $T = 303$ K.

$$\frac{PV}{RT} = 1 + \frac{A_1}{V} + \frac{A_2}{V^2}$$

P (atm)	0.985	1.108	1.363	1.631
V (mL)	25,000	22,200	18,000	15,000

15.28 Three disease-carrying organisms decay exponentially in lake water according to the following model:

$$p(t) = Ae^{-1.5t} + Ae^{-0.3t} + Ae^{-0.05t}$$

Estimate the initial population of each organism (A , B , and C) given the following measurements:

t , hr	0.5	1	2	3	4	5	6	7	9
$p(t)$	6.0	4.4	3.2	2.7	2.2	1.9	1.7	1.4	1.1

PROBLEMS

16.1 The pH in a reactor varies sinusoidally over the course of a day. Use least-squares regression to fit Eq. (16.11) to the following data. Use your fit to determine the mean, amplitude, and time of maximum pH. Note that the period is 24 hr

Time, hr	0	2	4	5	7	9
pH	7.6	7.2	7	6.5	7.5	7.2
Time, hr	12	15	20	22	24	
pH	8.9	9.1	8.9	7.9	7	

16.2 The solar radiation for Tucson, Arizona, has been tabulated as

Time, mo	J	F	M	A	M	J
Radiation, W/m ²	144	188	245	311	351	359
Time, mo	J	A	S	O	N	D
Radiation, W/m ²	308	287	260	211	159	131

Assuming each month is 30 days long, fit a sinusoid to these data. Use the resulting equation to predict the radiation in mid-August.

from $t = 0$ to 2π . Add a random component to the signal with the function `randn`. Use `fft` to generate a DFT of these values and plot the results.

16.11 Use MATLAB to generate 32 points for the sinusoid depicted in Fig. 16.2 from $t = 0$ to 6 s. Compute the DFT and create subplots of (a) the original signal, (b) the real part, and (c) the imaginary part of the DFT versus frequency.

16.12 Use the `fft` function to compute a DFT for the triangular wave from Prob. 16.6. Sample the wave from $t = 0$ to $4T$ using 128 sample points.

16.13 Develop an M-file function that uses the `fft` function to generate a power spectrum plot. Use it to solve Prob. 16.9.

16.3 The average values of a function can be determined by

$$\bar{f} = \frac{\int_0^T f(t) dt}{T}$$

Use this relationship to verify the results of Eq. (16.13).

16.4 In electric circuits, it is common to see current behavior in the form of a square wave as shown in Fig. P16.4 (notice that square wave differs from the one described in Example 16.2). Solving for the Fourier series from

$$f(t) = \begin{cases} A_0 & 0 \leq t \leq T/2 \\ -A_0 & T/2 \leq t \leq T \end{cases}$$

the Fourier series can be represented as

$$f(t) = \sum_{n=1}^{\infty} \left(\frac{4A_0}{(2n-1)\pi} \right) \sin \left(\frac{2\pi(2n-1)t}{T} \right)$$

Develop a MATLAB function to generate a plot of the first n terms of the Fourier series individually, as well as the sum of these six terms. Design your function so that it plots the curves from $t = 0$ to $4T$. Use thin dotted red lines for the individual terms and a bold black solid line for the summation (i.e., 'k-', 'linewidth', 2). The function's first line should be

```
function [t,f] = FourierSquare(A0,T,n)
```

Let $A_0 = 1$ and $T = 0.25$ s.

16.5 Use a continuous Fourier series to approximate the sawtooth wave in Fig. P16.5. Plot the first four terms along with the summation. In addition, construct amplitude and phase line spectra for the first four terms.

16.6 Use a continuous Fourier series to approximate the triangular wave form in Fig. P16.6. Plot the first four terms along with the summation. In addition, construct amplitude and phase line spectra for the first four terms.

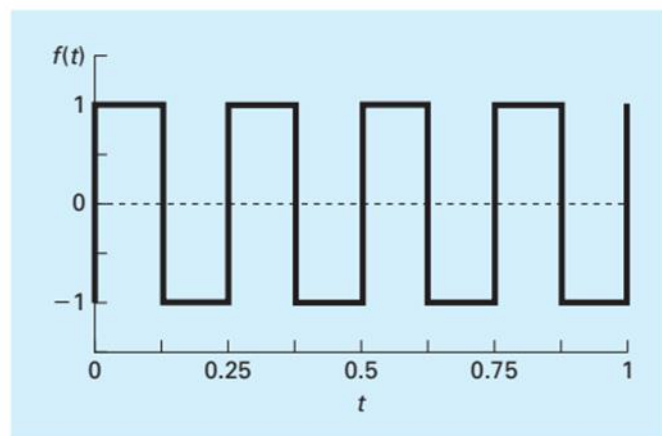


FIGURE P16.4

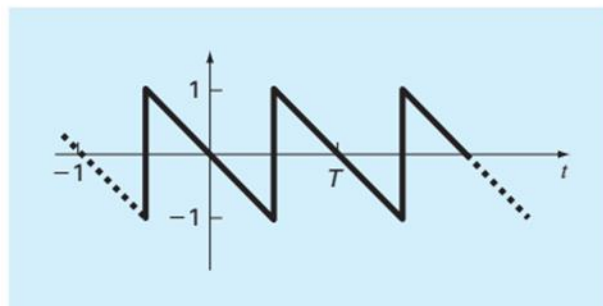


FIGURE P16.5

A sawtooth wave.

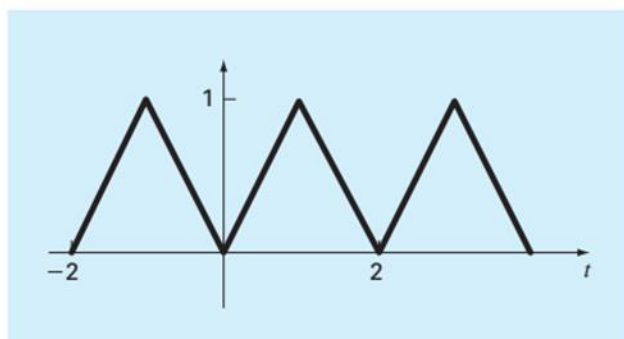


FIGURE P16.6

A triangular wave.

16.7 Use the *Maclaurin series expansions* for e^x , $\cos x$ and $\sin x$ to prove Euler's formula (Eq. 16.21).

16.8 A half-wave rectifier can be characterized by

$$C_1 = \left[\frac{1}{\pi} + \frac{1}{2} \sin t - \frac{2}{3\pi} \cos 2t - \frac{2}{15\pi} \cos 4t - \frac{2}{35\pi} \cos 6t - \dots \right]$$

where C_1 is the amplitude of the wave.

(a) Plot the first four terms along with the summation.

(b) Construct amplitude and phase line spectra for the first four terms.

16.9 Duplicate Example 16.3, but for 64 points sampled at a rate of $\Delta t = 0.01$ s from the function

$$f(t) = \cos[2\pi(12.5)t] + \cos[2\pi(25)t]$$

Use `fft` to generate a DFT of these values and plot the results.

16.10 Use MATLAB to generate 64 points from the function

$$f(t) = \cos(10t) + \sin(3t)$$

17.3 Use Newton's interpolating polynomial to determine y at $x = 8$ to the best possible accuracy. Compute the finite divided differences as in Fig. 17.5, and order your points to attain optimal accuracy and convergence. That is, the points should be centered around and as close as possible to the unknown.

x	0	1	2	5.5	11	13	16	18
y	0.5	3.134	5.3	9.9	10.2	9.35	7.2	6.2

17.4 Given the data

x	1	2	2.5	3	4	5
$f(x)$	0	5	6.5	7	3	1

(a) Calculate $f(3.4)$ using Newton's interpolating polynomials of order 1 through 3. Choose the sequence of the points for your estimates to attain the best possible accuracy. That is, the points should be centered around and as close as possible to the unknown.

(b) Repeat (a) but use the Lagrange polynomial.

17.5 Given the data

x	1	2	3	5	6
$f(x)$	7	4	5.5	40	82

Calculate $f(4)$ using Newton's interpolating polynomials of order 1 through 4. Choose your base points to attain good accuracy. That is, the points should be centered around and as close as possible to the unknown. What do your results indicate regarding the order of the polynomial used to generate the data in the table?

17.6 Repeat Prob. 17.5 using the Lagrange polynomial of order 1 through 3.

17.7 Table P15.5 lists values for dissolved oxygen concentration in water as a function of temperature and chloride concentration.

(a) Use quadratic and cubic interpolation to determine the oxygen concentration for $T = 12^\circ\text{C}$ and $c = 10\text{ g/L}$.

(b) Use linear interpolation to determine the oxygen concentration for $T = 12^\circ\text{C}$ and $c = 15\text{ g/L}$.

(c) Repeat (b) but use quadratic interpolation.

17.8 Employ inverse interpolation using a cubic interpolating polynomial and bisection to determine the value of x that corresponds to $f(x) = 1.7$ for the following tabulated data:

x	1	2	3	4	5	6	7
$f(x)$	3.6	1.8	1.2	0.9	0.72	1.5	0.51429

17.9 Employ inverse interpolation to determine the value of x that corresponds to $f(x) = 0.93$ for the following tabulated data:

x	0	1	2	3	4	5
$f(x)$	0	0.5	0.8	0.9	0.941176	0.961538

Note that the values in the table were generated with the function $f(x) = x^2/(1+x^2)$.

(a) Determine the correct value analytically.

(b) Use quadratic interpolation and the quadratic formula to determine the value numerically.

(c) Use cubic interpolation and bisection to determine the value numerically.

17.10 Use the portion of the given steam table for superheated water at 200 MPa to find (a) the corresponding entropy s for a specific volume v of 0.118 with linear interpolation, (b) the same corresponding entropy using quadratic interpolation, and (c) the volume corresponding to an entropy of 6.45 using inverse interpolation.

$v, \text{m}^3/\text{kg}$	0.10377	0.11144	0.12547
$s, \text{kJ}/(\text{kg K})$	6.4147	6.5453	6.7664

17.11 The following data for the density of nitrogen gas versus temperature come from a table that was measured with high precision. Use first- through fifth-order polynomials to estimate the density at a temperature of 330 K. What is your best estimate? Employ this best estimate and inverse interpolation to determine the corresponding temperature.

T, K	200	250	300	350	400	450
Density, kg/m^3	1.708	1.367	1.139	0.967	0.854	0.759

17.12 Ohm's law states that the voltage drop V across an ideal resistor is linearly proportional to the current i flowing through the resistor as in $V = iR$, where R is the resistance. However, real resistors may not always obey Ohm's law. Suppose that you performed some very precise experiments to measure the voltage drop and corresponding current for a resistor. The following results suggest a curvilinear relationship rather than the straight line represented by Ohm's law:

i	-1	-0.5	-0.25	0.25	0.5	1
V	-637	-96.5	-20.5	20.5	96.5	637

To quantify this relationship, a curve must be fit to the data. Because of measurement error, regression would typically be the preferred method of curve fitting for analyzing such experimental data. However, the smoothness of the relationship, as well as the precision of the experimental methods, suggests that interpolation might be appropriate. Use a fifth-order interpolating polynomial to fit the data and compute V for $i = 0.10$.

17.13 Bessel functions often arise in advanced engineering analyses such as the study of electric fields. Here are some selected values for the zero-order Bessel function of the first kind

x	1.8	2.0	2.2	2.4	2.6
$J_1(x)$	0.5815	0.5767	0.5560	0.5202	0.4708

Estimate $J_1(2.1)$ using third- and fourth-order interpolating polynomials. Determine the percent relative error for each case based on the true value, which can be determined with MATLAB's built-in function `besselj`.

17.14 Repeat Example 17.6 but using first-, second-, third-, and fourth-order interpolating polynomials to predict the population in 2000 based on the most recent data. That is, for the linear prediction use the data from 1980 and 1990, for the quadratic prediction use the data from 1970, 1980, and 1990, and so on. Which approach yields the best result?

17.15 The specific volume of a superheated steam is listed in steam tables for various temperatures. For example, at a pressure of 3000 lb/in², absolute:

$T, ^\circ\text{C}$	370	382	394	406	418
$v, \text{ft}^3/\text{kg}$	5.9313	7.5838	8.8428	9.796	10.5311

Determine v at $T = 750^\circ\text{F}$.

17.16 The vertical stress σ_z under the corner of a rectangular area subjected to a uniform load of intensity q is given by the solution of Boussinesq's equation:

$$\sigma = \frac{q}{4\pi} \left[\frac{2mn\sqrt{m^2 + n^2 + 1}}{m^2 + n^2 + 1 + m^2n^2} \frac{m^2 + n^2 + 2}{m^2 + n^2 + 1} + \sin^{-1} \left(\frac{2mn\sqrt{m^2 + n^2 + 1}}{m^2 + n^2 + 1 + m^2n^2} \right) \right]$$

Because this equation is inconvenient to solve manually, it has been reformulated as

$$\sigma_z = q f_z(m, n)$$

where $f_z(m, n)$ is called the influence value, and m and n are dimensionless ratios, with $m = a/z$ and $n = b/z$ and a and b are defined in Fig. P17.16. The influence value is then tabulated, a portion of which is given in Table P17.16. If $a = 4.6$ and $b = 14$, use a third-order interpolating polynomial to compute σ_z at a depth 10 m below the corner of a rectangular footing that is subject to a total load of 100 t

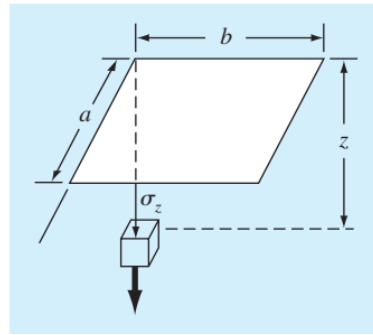


FIGURE P17.16

(metric tons). Express your answer in tonnes per square meter. Note that q is equal to the load per area.

TABLE P17.16

m	$n = 1.2$	$n = 1.4$	$n = 1.6$
0.1	0.02926	0.03007	0.03058
0.2	0.05733	0.05894	0.05994
0.3	0.08323	0.08561	0.08709
0.4	0.10631	0.10941	0.11135
0.5	0.12626	0.13003	0.13241
0.6	0.14309	0.14749	0.15027
0.7	0.15703	0.16199	0.16515
0.8	0.16843	0.17389	0.17739

17.17 You measure the voltage drop V across a resistor for a number of different values of current i . The results are

i	0.25	0.75	1.25	1.5	2.0
V	-0.45	-0.6	0.70	1.88	6.0

Use first- through fourth-order polynomial interpolation to estimate the voltage drop for $i = 1.15$. Interpret your results.

17.18 The current in a wire is measured with great precision as a function of time:

t	0	0.1250	0.2500	0.3750	0.5000
i	0	6.24	7.75	4.85	0.0000

Determine i at $t = 0.23$.

17.19 The acceleration due to gravity at an altitude y above the surface of the earth is given by

y, m	0	30,000	60,000	90,000	120,000
$g, \text{m/s}^2$	9.8100	9.7487	9.6879	9.6278	9.5682

Compute g at $y = 55,000$ m.

PROBLEMS

17.1 The following data come from a table that was measured with high precision. Use the best numerical method (for this type of problem) to determine y at $x = 3.5$. Note that a polynomial will yield an exact value. Your solution should prove that your result is exact.

x	0	1.8	5	6	8.2	9.2	12
y	26	16.415	5.375	3.5	2.015	2.54	8

17.2 Use Newton's interpolating polynomial to determine y at $x = 3.5$ to the best possible accuracy. Compute the finite divided differences as in Fig. 17.5, and order your points to attain optimal accuracy and convergence. That is, the points should be centered around and as close as possible to the unknown.

x	0	1	2.5	3	4.5	5	6
y	2	5.4375	7.3516	7.5625	8.4453	9.1875	12

17.20 Temperatures are measured at various points on a heated plate (Table P17.20). Estimate the temperature at (a) $x = 4$, $y = 3.2$, and (b) $x = 4.3$, $y = 2.7$.

TABLE P17.20 Temperatures ($^{\circ}\text{C}$) at various points on a square heated plate.

	$x = 0$	$x = 2$	$x = 4$	$x = 6$	$x = 8$
$y = 0$	100.00	90.00	80.00	70.00	60.00
$y = 2$	85.00	64.49	53.50	48.15	50.00
$y = 4$	70.00	48.90	38.43	35.03	40.00
$y = 6$	55.00	38.78	30.39	27.07	30.00
$y = 8$	40.00	35.00	30.00	25.00	20.00

17.21 Use the portion of the given steam table for superheated H_2O at 200 MPa to (a) find the corresponding entropy s for a specific volume v of $0.108 \text{ m}^3/\text{kg}$ with linear interpolation, (b) find the same corresponding entropy using quadratic interpolation, and (c) find the volume corresponding to an entropy of 6.6 using inverse interpolation.

$v \text{ (m}^3/\text{kg)}$	0.10377	0.11144	0.12540
$s \text{ (kJ/kg} \cdot \text{K)}$	6.4147	6.5453	6.7664

PROBLEMS

18.1 Given the data

x	1	2	2.5	3	4	5
$f(x)$	1	5	7	8	2	1

Fit these data with (a) a cubic spline with natural end conditions, (b) a cubic spline with not-a-knot end conditions, and (c) piecewise cubic Hermite interpolation.

18.2 A reactor is thermally stratified as in the following table:

Depth, m	0	0.5	1	1.5	2	2.5	3
Temperature, °C	70	70	55	22	13	10	10

Based on these temperatures, the tank can be idealized as two zones separated by a strong temperature gradient or *thermocline*. The depth of the thermocline can be defined as the inflection point of the temperature-depth curve—that is, the point at which $d^2T/dz^2 = 0$. At this depth, the heat flux from the surface to the bottom layer can be computed with Fourier's law:

$$J = -k \frac{dT}{dz}$$

Use a clamped cubic spline fit with zero end derivatives to determine the thermocline depth. If $k = 0.01 \text{ cal}/(\text{s} \cdot \text{cm} \cdot ^\circ\text{C})$ compute the flux across this interface.

18.3 The following is the built-in `humps` function that MATLAB uses to demonstrate some of its numerical capabilities:

$$f(x) = \frac{1}{(x - 0.3)^2 + 0.01} + \frac{1}{(x - 0.9)^2 + 0.04} - 6$$

The `humps` function exhibits both flat and steep regions over a relatively short x range. Here are some values that have been generated at intervals of 0.1 over the range from $x = 0$ to 1:

x	0	0.1	0.2	0.3	0.4	0.5
$f(x)$	5.176	15.471	45.887	96.500	47.448	19.000
x	0.6	0.7	0.8	0.9	1	
$f(x)$	11.692	12.382	17.846	21.703	16.000	

Fit these data with a (a) cubic spline with not-a-knot end conditions and (b) piecewise cubic Hermite interpolation. In both cases, create a plot comparing the fit with the exact `humps` function.

18.4 Develop a plot of a cubic spline fit of the following data with (a) natural end conditions and (b) not-a-knot end

conditions. In addition, develop a plot using (c) piecewise cubic Hermite interpolation.

x	0	100	200	400
$f(x)$	0	0.82436	1.00000	0.73576
x	600	800	1000	
$f(x)$	0.40601	0.19915	0.09158	

In each case, compare your plot with the following equation which was used to generate the data:

$$f(x) = \frac{x}{200} e^{-x/200+1}$$

18.5 The following data are sampled from the step function depicted in Fig. 18.1:

x	-1	-0.6	-0.2	0.2	0.6	1
$f(x)$	0	0	0	1	1	1

Fit these data with a (a) cubic spline with not-a-knot end conditions, (b) cubic spline with zero-slope clamped end conditions, and (c) piecewise cubic Hermite interpolation. In each case, create a plot comparing the fit with the step function. **18.6** Develop an M-file to compute a cubic spline fit with natural end conditions. Test your code by using it to duplicate Example 18.3.

18.7 The following data were generated with the fifth-order polynomial: $f(x) = 0.0185x^5 - 0.444x^4 + 3.9125x^3 - 15.456x^2 + 27.069x - 14.1$:

x	1	3	5	6	7	9
$f(x)$	1.000	2.172	4.220	5.430	4.912	9.120

(a) Fit these data with a cubic spline with not-a-knot end conditions. Create a plot comparing the fit with the function. (b) Repeat (a) but use clamped end conditions where the end slopes are set at the exact values as determined by differentiating the function.

18.8 Bessel functions often arise in advanced engineering and scientific analyses such as the study of electric fields. These functions are usually not amenable to straightforward evaluation and, therefore, are often compiled in standard mathematical tables. For example,

x	1.8	2	2.2	2.4	2.6
$J_1(x)$	0.5815	0.5767	0.556	0.5202	0.4708

Estimate $J_1(2.1)$, (a) using an interpolating polynomial and (b) using cubic splines. Note that the true value is 0.5683. **18.9** The following data define the sea-level concentration of dissolved oxygen for fresh water as a function of temperature:

$T, ^\circ\text{C}$	0	8	16	24	32	40
$o, \text{mg/L}$	14.621	11.843	9.870	8.418	7.305	6.413

Use MATLAB to fit the data with (a) piecewise linear interpolation, (b) a fifth-order polynomial, and (c) a spline. Display the results graphically and use each approach to estimate $o(27)$. Note that the exact result is 7.986 mg/L.

18.10 (a) Use MATLAB to fit a cubic spline to the following data to determine y at $x = 1.5$:

x	0	2	4	7	10	12
y	20	20	12	7	6	6

(b) Repeat (a), but with zero first derivatives at the end knots.

18.11 Runge's function is written as

$$f(x) = \frac{1}{1 + 25x^2}$$

Generate five equidistantly spaced values of this function over the interval: $[-1, 1]$. Fit these data with (a) a fourth-order polynomial, (b) a linear spline, and (c) a cubic spline. Present your results graphically.

18.12 Use MATLAB to generate eight points from the function

$$f(t) = \sin^2 t$$

from $t = 0$ to 2π . Fit these data using (a) cubic spline with not-a-knot end conditions, (b) cubic spline with derivative end conditions equal to the exact values calculated with differentiation, and (c) piecewise cubic hermite interpolation. Develop plots of each fit as well as plots of the absolute error ($E_t = \text{approximation} - \text{true}$) for each.

18.13 The drag coefficient for spheres such as sporting balls is known to vary as a function of the *Reynolds* number Re , a dimensionless number that gives a measure of the ratio of inertial forces to viscous forces:

$$\text{Re} = \frac{\rho V D}{\mu}$$

where ρ = the fluid's density (kg/m^3), V = its velocity (m/s), D = diameter (m), and μ = dynamic viscosity ($\text{N} \cdot \text{s/m}^2$).

Although the relationship of drag to the Reynolds number is sometimes available in equation form, it is frequently tabulated. For example, the following table provides values for a smooth spherical ball:

Re ($\times 10^{-4}$)	2	5.8	16.8	27.2	29.9	33.9	
C_D	0.52	0.52	0.52	0.5	0.49	0.44	
Re ($\times 10^{-4}$)	36.3	40	46	60	100	200	400
C_D	0.18	0.074	0.067	0.08	0.12	0.16	0.19

(a) Develop a MATLAB function that employs the `spline` function to return a value of C_D as a function of the Reynolds number. The first line of the function should be

```
function CDout = Drag(ReCD,ReIn)
```

where `ReCD` = a 2-row matrix containing the table, `ReIn` = the Reynolds number at which you want to estimate the drag, and `CDout` = the corresponding drag coefficient.

(b) Write a script that uses the function developed in part (a) to generate a labeled plot of the drag force versus velocity (recall Sec. 1.4). Use the following parameter values for the script: $D = 22$ cm, $\rho = 1.3$ kg/m³, and $\mu = 1.78 \times 10^{-5}$ Pa · s. Employ a range of velocities from 4 to 40 m/s for your plot.

18.14 The following function describes the temperature distribution on a rectangular plate for the range $-2 \leq x \leq 0$ and $0 \leq y \leq 3$

$$T = 2 + x - y + 2x^2 + 2xy + y^2$$

Develop a script to: (a) Generate a meshplot of this function using the MATLAB function `surf`. Employ the `linspace` function with default spacing (i.e., 100 interior points) to generate the x and y values. (b) Use the MATLAB function `interp2` with the default interpolation option ('linear') to compute the temperature at $x = -1.63$ and $y = 1.627$. Determine the percent relative error of your result. (c) Repeat (b), but with 'spline'. Note: for parts (b) and (c), employ the `linspace` function with 9 interior points.

PROBLEMS

19.1 Derive Eq. (19.4) by integrating Eq. (19.3).

19.2 Evaluate the following integral:

$$\int_0^4 (1 - e^{-x}) dx$$

(a) analytically, (b) single application of the trapezoidal rule, (c) composite trapezoidal rule with $n = 2$ and 4, (d) single application of Simpson's 1/3 rule, (e) composite Simpson's

1/3 rule with $n = 4$, (f) Simpson's 3/8 rule, and (g) composite Simpson's rule, with $n = 5$. For each of the numerical estimates (b) through (g), determine the true percent relative error based on (a).

19.3 Evaluate the following integral:

$$\int_0^{\pi/2} (8 + 4 \cos x) dx$$

(a) analytically, (b) single application of the trapezoidal rule, (c) composite trapezoidal rule with $n = 2$ and 4, (d) single application of Simpson's 1/3 rule, (e) composite Simpson's 1/3 rule with $n = 4$, (f) Simpson's 3/8 rule, and (g) composite Simpson's rule, with $n = 5$. For each of the numerical estimates (b) through (g), determine the true percent relative error based on (a).

19.4 Evaluate the following integral:

$$\int_{-2}^4 (1 - x - 4x^3 + 2x^5) dx$$

(a) analytically, (b) single application of the trapezoidal rule, (c) composite trapezoidal rule with $n = 2$ and 4, (d) single application of Simpson's 1/3 rule, (e) Simpson's 3/8 rule, and (f) Boole's rule. For each of the numerical estimates (b) through (f), determine the true percent relative error based on (a).

19.5 The function

$$f(x) = e^{-x}$$

can be used to generate the following table of unequally spaced data:

x	0	0.1	0.3	0.5	0.7	0.95	1.2
$f(x)$	1	0.9048	0.7408	0.6065	0.4966	0.3867	0.3012

Evaluate the integral from $a = 0$ to $b = 0.6$ using (a) analytical means, (b) the trapezoidal rule, and (c) a combination of the trapezoidal and Simpson's rules wherever possible to attain the highest accuracy. For (b) and (c), compute the true percent relative error.

19.6 Evaluate the double integral

$$\int_{-2}^2 \int_0^4 (x^2 - 3y^2 + xy^3) dx dy$$

(a) analytically, (b) using the composite trapezoidal rule with $n = 2$, and (c) using single applications of Simpson's 1/3 rule. For (b) and (c), compute the percent relative error.

19.7 Evaluate the triple integral

$$\int_{-4}^4 \int_0^6 \int_{-1}^3 (x^3 - 2yz) dx dy dz$$

(a) analytically, and (b) using single applications of Simpson's 1/3 rule. For (b), compute the true percent relative error.

19.8 Determine the distance traveled from the following velocity data:

t	1	2	3.25	4.5	6	7	8	8.5	9	10
v	5	6	5.5	7	8.5	8	6	7	7	5

(a) Use the trapezoidal rule. In addition, determine the average velocity.
(b) Fit the data with a cubic equation using polynomial regression. Integrate the cubic equation to determine the distance.

19.9 Water exerts pressure on the upstream face of a dam as shown in Fig. P19.9. The pressure can be characterized by

$$p(z) = \rho g(D - z)$$

where $p(z)$ = pressure in pascals (or N/m^2) exerted at an elevation z meters above the reservoir bottom; ρ = density of water, which for this problem is assumed to be a constant 10^3 kg/m^3 ; g = acceleration due to gravity (9.81 m/s^2); and D = elevation (in m) of the water surface above the reservoir bottom. According to Eq. (P19.9), pressure increases linearly with depth, as depicted in Fig. P19.9a. Omitting atmospheric pressure (because it works against both sides of the dam face and essentially cancels out), the total force f_t can be determined by multiplying pressure times the area of the dam face (as shown in Fig. P19.9b). Because both pressure and area vary with elevation, the total force is obtained by evaluating

$$f_t = \int_0^D \rho g w(z)(D - z) dz$$

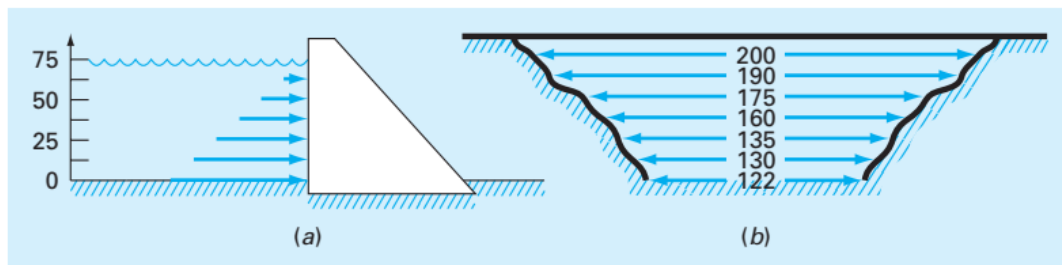


FIGURE P19.9

Water exerting pressure on the upstream face of a dam: (a) side view showing force increasing linearly with depth; (b) front view showing width of dam in meters.

where $w(z)$ = width of the dam face (m) at elevation z (Fig. P19.9b). The line of action can also be obtained by evaluating

$$d = \frac{\int_0^D \rho g z w(z)(D - z) dz}{\int_0^D \rho g w(z)(D - z) dz}$$

Use Simpson's rule to compute f_t and d .

19.10 The force on a sailboat mast can be represented by the following function:

$$f(z) = 200 \left(\frac{z}{5 + z} \right) e^{-2z/H}$$

where z = the elevation above the deck and H = the height of the mast. The total force F exerted on the mast can be determined by integrating this function over the height of the mast:

$$F = \int_0^H f(z) dz$$

The line of action can also be determined by integration:

$$d = \frac{\int_0^H z f(z) dz}{\int_0^H f(z) dz}$$

(a) Use the composite trapezoidal rule to compute F and d for the case where $H = 30$ ($n = 6$).

(b) Repeat (a), but use the composite Simpson's 1/3 rule.

19.11 A wind force distributed against the side of a skyscraper is measured as

Height l , m	0	30	60	90	120
Force, $F(l)$, N/m	0	340	1200	1550	2700

Height l , m	150	180	210	240
Force, $F(l)$, N/m	3100	3200	3500	3750

Compute the net force and the line of action due to this distributed wind.

19.12 An 11-m beam is subjected to a load, and the shear force follows the equation

$$V(x) = 5 + 0.25x^2$$

where V is the shear force, and x is length in distance along the beam. We know that $V = dM/dx$, and M is the bending moment. Integration yields the relationship

$$M = M_o + \int_0^x V dx$$

If M_o is zero and $x = 11$, calculate M using (a) analytical integration, (b) composite trapezoidal rule, and (c) composite Simpson's rules. For (b) and (c) use 1-m increments.

19.13 The total mass of a variable density rod is given by

$$m = \int_0^L \rho(x) A_c(x) dx$$

where m = mass, $\rho(x)$ = density, $A_c(x)$ = cross-sectional area, x = distance along the rod and L = the total length of the rod. The following data have been measured for a 20-m length rod. Determine the mass in grams to the best possible accuracy.

x , m	0	4	6	8	12	16	20
ρ , g/cm ³	4.00	3.95	3.89	3.80	3.60	3.41	3.30
A_c , cm ²	100	103	106	110	120	133	150

19.14 A transportation engineering study requires that you determine the number of cars that pass through an intersection traveling during morning rush hour. You stand at the side of the road and count the number of cars that pass every 4 minutes at several times as tabulated below. Use the best numerical method to determine (a) the total number of cars that pass between 7:30 and 9:15, and (b) the rate of cars going through the intersection per minute. (Hint: Be careful with units.)

Time (hr)	7:30	7:45	8:00	8:15	8:45	9:15
Rate (cars per 4 min)	18	23	14	24	20	9

19.15 Determine the average value for the data in Fig. P19.15. Perform the integral needed for the average in the order shown by the following equation:

$$I = \int_{x_0}^{x_n} \left[\int_{y_0}^{y_m} f(x, y) dy \right] dx$$

19.16 Integration provides a means to compute how much mass enters or leaves a reactor over a specified time period, as in

$$M = \int_{t_1}^{t_2} Qc dt$$

where t_1 and t_2 = the initial and final times, respectively. This formula makes intuitive sense if you recall the analogy between integration and summation. Thus, the integral represents the summation of the product of flow times

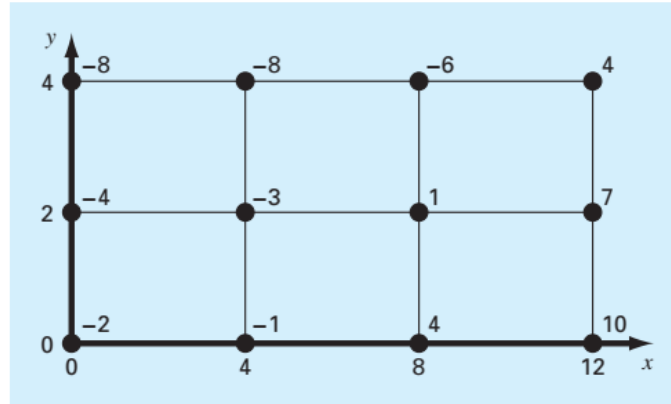


FIGURE P19.15

concentration to give the total mass entering or leaving from t_1 to t_2 . Use numerical integration to evaluate this equation for the data listed below:

$t, \text{ min}$	0	10	20	30	35	40	45	50
$Q, \text{ m}^3/\text{min}$	4	4.8	5.2	5.0	4.6	4.3	4.3	5.0
$c, \text{ mg/m}^3$	10	35	55	52	40	37	32	34

19.17 The cross-sectional area of a channel can be computed as

$$A_c = \int_0^B H(y) dy$$

where B = the total channel width (m), H = the depth (m), and y = distance from the bank (m). In a similar fashion, the average flow Q (m^3/s) can be computed as

$$Q = \int_0^B U(y)H(y) dy$$

where U = water velocity (m/s). Use these relationships and a numerical method to determine A_c and Q for the following data:

$y, \text{ m}$	0	2	4	5	6	9
$H, \text{ m}$	0.5	1.3	1.25	1.8	1	0.25
$U, \text{ m/s}$	0.03	0.06	0.05	0.13	0.11	0.02

19.18 The average concentration of a substance $\bar{c}$ (g/m^3) in a lake where the area A_s (m^2) varies with depth z (m) can be computed by integration:

$$\bar{c} = \frac{\int_0^Z c(z)A_s(z) dz}{\int_0^Z A_s(z) dz}$$

where Z = the total depth (m). Determine the average concentration based on the following data:

$z, \text{ m}$	0	4	8	12	16
$A, 10^6 \text{ m}^2$	9.8175	5.1051	1.9635	0.3927	0.0000
$c, \text{ g/m}^3$	10.2	8.5	7.4	5.2	4.1

19.19 As was done in Section 19.9, determine the work performed if a constant force of 1 N applied at an angle θ results in the following displacements. Use the `cumtrapz` function to determine the cumulative work and plot the result versus θ .

$x, \text{ m}$	0	1	2.8	3.9	3.8	3.2	1.3
$\theta, \text{ deg}$	0	30	60	90	120	150	180

19.20 Compute work as described in Sec. 19.9, but use the following equations for $F(x)$ and $\theta(x)$:

$$F(x) = 1.6x - 0.045x^2$$

$$\theta(x) = -0.00055x^3 + 0.0123x^2 + 0.13x$$

The force is in Newtons and the angle is in radians. Perform the integration from $x = 0$ to 30 m.

PROBLEMS

20.1 Use Romberg integration to evaluate

$$I = \int_1^2 \left(x + \frac{1}{x} \right)^2 dx$$

to an accuracy of $\varepsilon_s = 0.5\%$. Your results should be presented in the format of Fig. 20.1. Use the analytical solution of the integral to determine the percent relative error of the result obtained with Romberg integration. Check that ε_t is less than ε_s .

20.2 Evaluate the following integral (a) analytically, (b) Romberg integration ($\varepsilon_s = 0.5\%$), (c) the three-point

Gauss quadrature formula, and (d) MATLAB quad function:

$$I = \int_0^8 -0.055x^4 + 0.86x^3 - 4.2x^2 + 6.3x + 2 dx$$

20.3 Evaluate the following integral with (a) Romberg integration ($\varepsilon_s = 0.5\%$), (b) the two-point Gauss quadrature formula, and (c) MATLAB quad and quadl functions:

$$I = \int_0^3 xe^{2x} dx$$

19.21 As specified in the following table, a manufactured spherical particle has a density that varies as a function of the distance from its center ($r = 0$):

$r, \text{ mm}$	0	0.12	0.24	0.36	0.49
$\rho(\text{g/cm}^3)$	6	5.81	5.14	4.29	3.39
$r, \text{ mm}$	0.62	0.79	0.86	0.93	1
$\rho(\text{g/cm}^3)$	2.7	2.19	2.1	2.04	2

Use numerical integration to estimate the particle's mass (in g) and average density (in g/cm^3).

19.22 As specified in the following table, the earth's density varies as a function of the distance from its center ($r = 0$):

$r, \text{ km}$	0	1100	1500	2450	3400	3630
$\rho(\text{g/cm}^3)$	13	12.4	12	11.2	9.7	5.7
$r, \text{ km}$	4500	5380	6060	6280	6380	
$\rho(\text{g/cm}^3)$	5.2	4.7	3.6	3.4	3	

Use numerical integration to estimate the earth's mass (in metric tonnes) and average density (in g/cm^3). Develop vertically stacked subplots of (top) density versus radius, and (bottom) mass versus radius. Assume that the earth is a perfect sphere.

20.4 There is no closed form solution for the error function

$$\operatorname{erf}(a) = \frac{2}{\sqrt{\pi}} \int_0^a e^{-x^2} dx$$

Use the **(a)** two-point and **(b)** three-point Gauss-Legendre formulas to estimate $\operatorname{erf}(1.5)$. Determine the percent relative error for each case based on the true value, which can be determined with MATLAB's built-in function `erf`.

20.5 The force on a sailboat mast can be represented by the following function:

$$F = \int_0^H 200 \left(\frac{z}{5+z} \right) e^{-2z/H} dz$$

where z = the elevation above the deck and H = the height of the mast. Compute F for the case where $H = 30$ using **(a)** Romberg integration to a tolerance of $\varepsilon_s = 0.5\%$, **(b)** the two-point Gauss-Legendre formula, and **(c)** the MATLAB `quad` function.

20.6 The root-mean-square current can be computed as

$$I_{RMS} = \sqrt{\frac{1}{T} \int_0^T i^2(t) dt}$$

For $T = 1$, suppose that $i(t)$ is defined as

$$i(t) = 8e^{-t/T} \sin\left(2\pi \frac{t}{T}\right) \quad \text{for } 0 \leq t \leq T/2$$

$$i(t) = 0 \quad \text{for } T/2 \leq t \leq T$$

Evaluate the I_{RMS} using **(a)** Romberg integration to a tolerance of 0.1% , **(b)** the two- and three-point Gauss-Legendre formulas, and **(c)** the MATLAB `quad` function.

20.7 The heat required, ΔH (cal), to induce a temperature change, ΔT ($^{\circ}\text{C}$), of a material can be computed as

$$\Delta H = mC_p(T)\Delta T$$

where m = mass (g), and $C_p(T)$ = heat capacity [cal/(g $\cdot$ $^{\circ}\text{C}$)]. The heat capacity increases with temperature, T ($^{\circ}\text{C}$), according to

$$C_p(T) = 0.132 + 1.56 \times 10^{-4}T + 2.64 \times 10^{-7}T^2$$

Write a script that uses the `quad` function to generate a plot of ΔH versus ΔT for cases where $m = 1$ kg, the starting temperature is -100 $^{\circ}\text{C}$ and ΔT ranges from 0 to 300 $^{\circ}\text{C}$.

20.8 The amount of mass transported via a pipe over a period of time can be computed as

$$M = \int_{t_1}^{t_2} Q(t)c(t) dt$$

where M = mass (mg), t_1 = the initial time (min), t_2 = the final time (min), $Q(t)$ = flow rate (m^3/min), and $c(t)$ = concentration (mg/m^3). The following functional representations define the temporal variations in flow and concentration:

$$Q(t) = 9 + 5 \cos^2(0.4t)$$

$$c(t) = 5e^{-0.5t} + 2e^{0.15t}$$

Determine the mass transported between $t_1 = 2$ and $t_2 = 8$ min with **(a)** Romberg integration to a tolerance of 0.1% and **(b)** the MATLAB `quad` function.

20.9 Evaluate the double integral

$$\int_{-2}^2 \int_0^4 (x^2 - 3y^2 + xy^3) dx dy$$

(a) analytically and **(b)** using the MATLAB `dblquad` function. Use `help` to understand how to implement the function.

20.10 Compute work as described in Sec. 19.9, but use the following equations for $F(x)$ and $\theta(x)$:

$$F(x) = 1.6x - 0.045x^2$$

$$\theta(x) = -0.00055x^3 + 0.0123x^2 + 0.13x$$

The force is in newtons and the angle is in radians. Perform the integration from $x = 0$ to 30 m.

20.11 Perform the same computation as in Sec. 20.5, but for the current as specified by

$$i(t) = 6e^{-1.25t} \sin 2\pi t \quad \text{for } 0 \leq t \leq T/2$$

$$i(t) = 0 \quad \text{for } T/2 < t \leq T$$

where $T = 1$ s.

20.12 Compute the power absorbed by an element in a circuit as described in Sec. 20.5, but for a simple sinusoidal current $i = \sin(2\pi t/T)$ where $T = 1$ s.

(a) Assume that Ohm's law holds and $R = 5 \Omega$.

(b) Assume that Ohm's law does not hold and that voltage and current are related by the following nonlinear relationship: $V = (5i - 1.25i^3)$.

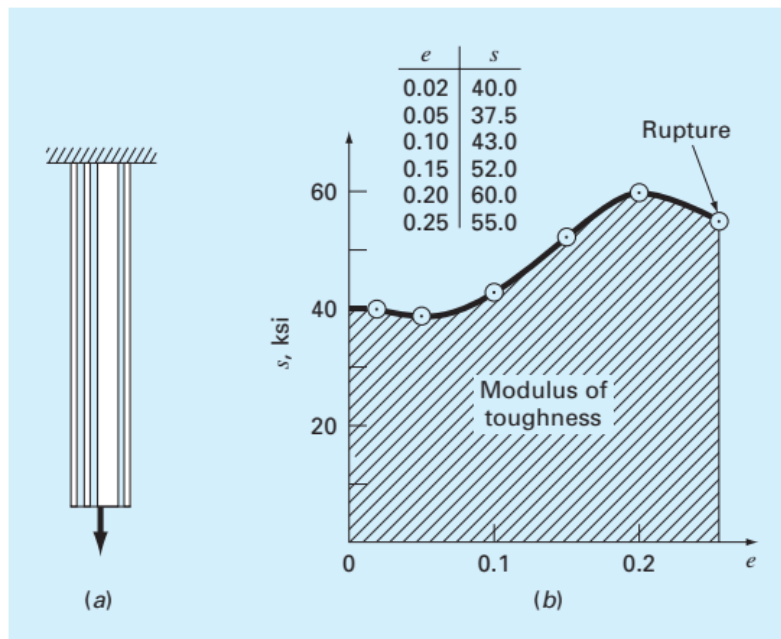
20.13 Suppose that the current through a resistor is described by the function

$$i(t) = (60 - t)^2 + (60 - t) \sin(\sqrt{t})$$

and the resistance is a function of the current:

$$R = 10i + 2i^{2/3}$$

Compute the average voltage over $t = 0$ to 60 using the composite Simpson's $1/3$ rule.

**FIGURE P20.16**

(a) A rod under axial loading and (b) the resulting stress-strain curve, where stress is in kips per square inch (10^3 lb/in²), and strain is dimensionless.

20.14 If a capacitor initially holds no charge, the voltage across it as a function of time can be computed as

$$V(t) = \frac{1}{C} \int_0^t i(t) dt$$

Use MATLAB to fit these data with a fifth-order polynomial. Then, use a numerical integration function along with a value of $C = 10^{-5}$ farad to generate a plot of voltage versus time.

t, s	0	0.2	0.4	0.6
$i, 10^{-3} A$	0.2	0.3683	0.3819	0.2282
t, s	0.8	1	1.2	
$i, 10^{-3} A$	0.0486	0.0082	0.1441	

20.15 The work done on an object is equal to the force times the distance moved in the direction of the force. The velocity of an object in the direction of a force is given by

$$\begin{aligned} v &= 4t & 0 \leq t \leq 5 \\ v &= 20 + (5 - t)^2 & 5 \leq t \leq 15 \end{aligned}$$

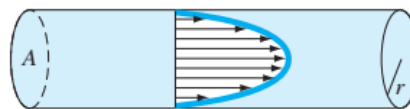
where v is in m/s. Determine the work if a constant force of 200 N is applied for all t .

20.16 A rod subject to an axial load (Fig. P20.16a) will be deformed, as shown in the stress-strain curve in Fig. P20.16b.

The area under the curve from zero stress out to the point of rupture is called the *modulus of toughness* of the material. It provides a measure of the energy per unit volume required to cause the material to rupture. As such, it is representative of the material's ability to withstand an impact load. Use numerical integration to compute the modulus of toughness for the stress-strain curve seen in Fig. P20.16b.

20.17 If the velocity distribution of a fluid flowing through a pipe is known (Fig. P20.17), the flow rate Q (i.e., the volume of water passing through the pipe per unit time) can be computed by $Q = \int v dA$, where v is the velocity, and A is the pipe's cross-sectional area. (To grasp the meaning of this relationship physically, recall the close connection between summation and integration.) For a circular pipe, $A = \pi r^2$ and $dA = 2\pi r dr$. Therefore,

$$Q = \int_0^r v(2\pi r) dr$$

**FIGURE P20.17**

where r is the radial distance measured outward from the center of the pipe. If the velocity distribution is given by

$$v = 2 \left(1 - \frac{r}{r_0} \right)^{1/6}$$

where r_0 is the total radius (in this case, 3 cm), compute Q using the composite trapezoidal rule. Discuss the results.
20.18 Using the following data, calculate the work done by stretching a spring that has a spring constant of $k = 300$ N/m to $x = 0.35$ m. To do this, first fit the data with a polynomial and then integrate the polynomial numerically to compute the work:

$F, 10^3 \cdot \text{N}$	0	0.01	0.028	0.046
x, m	0	0.05	0.10	0.15
$F, 10^3 \cdot \text{N}$	0.063	0.082	0.11	0.13
x, m	0.20	0.25	0.30	0.35

20.19 Evaluate the vertical distance traveled by a rocket if the vertical velocity is given by

$$\begin{aligned} v &= 11t^2 - 5t & 0 \leq t \leq 10 \\ v &= 1100 - 5t & 10 \leq t \leq 20 \\ v &= 50t + 2(t - 20)^2 & 20 \leq t \leq 30 \end{aligned}$$

20.20 The upward velocity of a rocket can be computed by the following formula:

$$v = u \ln \left(\frac{m_0}{m_0 - qt} \right) - gt$$

where v = upward velocity, u = velocity at which fuel is expelled relative to the rocket, m_0 = initial mass of the rocket at time $t = 0$, q = fuel consumption rate, and g = downward acceleration of gravity (assumed constant = 9.81 m/s^2). If $u = 1850 \text{ m/s}$, $m_0 = 160,000 \text{ kg}$, and $q = 2500 \text{ kg/s}$, determine how high the rocket will fly in 30 s.

20.21 The normal distribution is defined as

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

(a) Use MATLAB to integrate this function from $x = -1$ to 1 and from -2 to 2.

(b) Use MATLAB to determine the inflection points of this function.

20.22 Use Romberg integration to evaluate

$$\int_0^2 \frac{e^x \sin x}{1+x^2} dx$$

to an accuracy of $\varepsilon_s = 0.5\%$. Your results should be presented in the form of Fig. 20.1.

20.23 Recall that the velocity of the free-falling bungee jumper can be computed analytically as [(Eq. 1.9)]:

$$v(t) = \sqrt{\frac{gm}{c_d}} \tanh \left(\sqrt{\frac{gc_d}{m}} t \right)$$

where $v(t)$ = velocity (m/s), t = time (s), $g = 9.81 \text{ m/s}^2$, m = mass (kg), c_d = drag coefficient (kg/m).

(a) Use Romberg integration to compute how far the jumper travels during the first 8 seconds of free fall given $m = 80 \text{ kg}$ and $c_d = 0.2 \text{ kg/m}$. Compute the answer to $\varepsilon_s = 1\%$.
 (b) Perform the same computation with `quad`.

20.24 Prove that Eq. (20.30) is equivalent to Boole's rule.

20.25 As specified in the following table, the earth's density varies as a function of the distance from its center ($r = 0$):

r, km	0	1100	1500	2450	3400	3630	4500
$\rho, \text{g/cm}^3$	13	12.4	12	11.2	9.7	5.7	5.2
r, km	5380	6060	6280	6380			
$\rho, \text{g/cm}^3$	4.7	3.6	3.4	3			

Develop a script to fit these data with `interp1` using the `pchip` option. Generate a plot showing the resulting fit along with the data points. Then use one of MATLAB's integration functions to estimate the earth's mass (in metric tonnes) by integrating the output of the `interp1` function.

20.26 Develop an M-file function to implement Romberg integration based on Fig. 20.2. Test the function by using it to determine the integral of the polynomial from Example 20.1. Then use it to solve Prob. 20.1.

20.27 Develop an M-file function to implement adaptive quadrature based on Fig. 20.6. Test the function by using it to determine the integral of the polynomial from Example 20.1. Then use it to solve Prob. 20.20.

PROBLEMS

21.1 Compute forward and backward difference approximations of $O(h)$ and $O(h^2)$, and central difference approximations of $O(h^2)$ and $O(h^4)$ for the first derivative of $y = \cos x$ at $x = \pi/4$ using a value of $h = \pi/12$. Estimate the true percent relative error ε_t for each approximation.

21.2 Use centered difference approximations to estimate the first and second derivatives of $y = e^x$ at $x = 2$ for $h = 0.1$. Employ both $O(h^2)$ and $O(h^4)$ formulas for your estimates.

21.3 Use a Taylor series expansion to derive a centered finite-difference approximation to the third derivative that is second-order accurate. To do this, you will have to use four different expansions for the points x_{i-2} , x_{i-1} , x_{i+1} , and x_{i+2} . In each case, the expansion will be around the point x_i . The interval Δx will be used in each case of $i - 1$ and $i + 1$, and $2\Delta x$ will be used in each case of $i - 2$ and $i + 2$. The four equations must then be combined in a way to eliminate the first and second derivatives. Carry enough terms along in each expansion to evaluate the first term that will be truncated to determine the order of the approximation.

21.4 Use Richardson extrapolation to estimate the first derivative of $y = \cos x$ at $x = \pi/4$ using step sizes of $h_1 = \pi/3$ and $h_2 = \pi/6$. Employ centered differences of $O(h^2)$ for the initial estimates.

21.5 Repeat Prob. 21.4, but for the first derivative of $\ln x$ at $x = 5$ using $h_1 = 2$ and $h_2 = 1$.

21.6 Employ Eq. (21.21) to determine the first derivative of $y = 2x^4 - 6x^3 - 12x - 8$ at $x = 0$ based on values at $x_0 = -0.5$, $x_1 = 1$, and $x_2 = 2$. Compare this result with the true value and with an estimate obtained using a centered difference approximation based on $h = 1$.

21.7 Prove that for equispaced data points, Eq. (21.21) reduces to Eq. (4.25) at $x = x_1$.

21.8 Develop an M-file to apply a Romberg algorithm to estimate the derivative of a given function.

21.9 Develop an M-file to obtain first-derivative estimates for unequally spaced data. Test it with the following data:

x	0.6	1.5	1.6	2.5	3.5
$f(x)$	0.9036	0.3734	0.3261	0.08422	0.01596

where $f(x) = 5e^{-2x}$. Compare your results with the true derivatives.

21.10 Develop an M-file function that computes first and second derivative estimates of order $O(h^2)$ based on the formulas in Figs. 21.3 through 21.5. The function's first line should be set up as

```
function [dydx, d2ydx2] = diffeq(x,y)
```

where x and y are input vectors of length n containing the values of the independent and dependent variables, respectively, and $dydx$ and $d2ydx2$ are output vectors of length n containing the first- and second-derivative estimates at each value of the independent variable. The function should generate a plot of $dydx$ and $d2ydx2$ versus x . Have your M-file return an error message if (a) the input vectors are not the same length, or (b) the values for the independent variable are not equally spaced. Test your program with the data from Prob. 21.11.

21.11 The following data were collected for the distance traveled versus time for a rocket:

t, s	0	25	50	75	100	125
y, km	0	32	58	78	92	100

Use numerical differentiation to estimate the rocket's velocity and acceleration at each time.

21.12 A jet fighter's position on an aircraft carrier's runway was timed during landing:

$t, \text{ s}$	0	0.52	1.04	1.75	2.37	3.25	3.83
$x, \text{ m}$	153	185	208	249	261	271	273

where x is the distance from the end of the carrier. Estimate (a) velocity (dx/dt) and (b) acceleration (dv/dt) using numerical differentiation.

21.13 Use the following data to find the velocity and acceleration at $t = 10$ seconds:

Time, $t, \text{ s}$	0	2	4	6	8	10	12	14	16
Position, $x, \text{ m}$	0	0.7	1.8	3.4	5.1	6.3	7.3	8.0	8.4

Use second-order correct (a) centered finite-difference, (b) forward finite-difference, and (c) backward finite-difference methods.

21.14 A plane is being tracked by radar, and data are taken every second in polar coordinates θ and r :

$t, \text{ s}$	200	202	204	206	208	210
$\theta, (\text{rad})$	0.75	0.72	0.70	0.68	0.67	0.66
$r, \text{ m}$	5120	5370	5560	5800	6030	6240

At 206 seconds, use the centered finite-difference (second-order correct) to find the vector expressions for velocity $\vec{v}$ and acceleration $\vec{a}$. The velocity and acceleration given in polar coordinates are

$$\vec{v} = \dot{r}\vec{e}_r + r\dot{\theta}\vec{e}_\theta \quad \text{and} \quad \vec{a} = (\ddot{r} - r\dot{\theta}^2)\vec{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\vec{e}_\theta$$

21.15 Use regression to estimate the acceleration at each time for the following data with second-, third-, and fourth-order polynomials. Plot the results:

t	1	2	3.25	4.5	6	7	8	8.5	9.3	10
v	10	12	11	14	17	16	12	14	14	10

21.16 The normal distribution is defined as

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

Use MATLAB to determine the inflection points of this function.

21.17 The following data were generated from the normal distribution:

x	-2	-1.5	-1	-0.5	0
$f(x)$	0.05399	0.12952	0.24197	0.35207	0.39894

x	0.5	1	1.5	2
$f(x)$	0.35207	0.24197	0.12952	0.05399

Use MATLAB to estimate the inflection points of these data.

21.18 Use the `diff(y)` command to develop a MATLAB M-file function to compute finite-difference approximations to the first and second derivative at each x value in the table below. Use finite-difference approximations that are second-order correct, $O(x^2)$:

x	0	1	2	3	4	5	6	7	8	9	10
y	1.4	2.1	3.3	4.8	6.8	6.6	8.6	7.5	8.9	10.9	10

21.19 The objective of this problem is to compare second-order accurate forward, backward, and centered finite-difference approximations of the first derivative of a function to the actual value of the derivative. This will be done for

$$f(x) = e^{-2x} - x$$

- Use calculus to determine the correct value of the derivative at $x = 2$.
- Develop an M-file function to evaluate the centered finite-difference approximations, starting with $x = 0.5$. Thus, for the first evaluation, the x values for the centered difference approximation will be $x = 2 \pm 0.5$ or $x = 1.5$ and 2.5 . Then, decrease in increments of 0.1 down to a minimum value of $\Delta x = 0.01$.
- Repeat part (b) for the second-order forward and backward differences. (Note that these can be done at the same time that the centered difference is computed in the loop.)
- Plot the results of (b) and (c) versus x . Include the exact result on the plot for comparison.

21.20 You have to measure the flow rate of water through a small pipe. In order to do it, you place a bucket at the pipe's outlet and measure the volume in the bucket as a function of time as tabulated below. Estimate the flow rate at $t = 7 \text{ s}$.

Time, s	0	1	5	8
Volume, cm^3	0	1	8	16.4

21.21 The velocity v (m/s) of air flowing past a flat surface is measured at several distances y (m) away from the surface.

Use *Newton's viscosity law* to determine the shear stress τ (N/m²) at the surface ($y = 0$),

$$\tau = \mu \frac{du}{dy}$$

Assume a value of dynamic viscosity $\mu = 1.8 \times 10^{-5}$ N · s/m².

$y, \text{ m}$	0	0.002	0.006	0.012	0.018	0.024
$u, \text{ m/s}$	0	0.287	0.899	1.915	3.048	4.299

21.22 Fick's first diffusion law states that

$$\text{Mass ux} = -D \frac{dc}{dx} \quad (\text{P21.22})$$

where mass flux = the quantity of mass that passes across a unit area per unit time (g/cm²/s), D = a diffusion coefficient (cm²/s), c = concentration (g/cm³), and x = distance (cm). An environmental engineer measures the following concentration of a pollutant in the pore waters of sediments underlying a lake ($x = 0$ at the sediment-water interface and increases downward):

$x, \text{ cm}$	0	1	3
$c, 10^{-6} \text{ g/cm}^3$	0.06	0.32	0.6

Use the best numerical differentiation technique available to estimate the derivative at $x = 0$. Employ this estimate in conjunction with Eq. (P21.22) to compute the mass flux of pollutant out of the sediments and into the overlying waters ($D = 1.52 \times 10^{-6}$ cm²/s). For a lake with 3.6×10^6 m² of sediments, how much pollutant would be transported into the lake over a year's time?

21.23 The following data were collected when a large oil tanker was loading:

$t, \text{ min}$	0	10	20	30	45	60	75
$V, 10^6 \text{ barrels}$	0.4	0.7	0.77	0.88	1.05	1.17	1.35

Calculate the flow rate Q (i.e., dV/dt) for each time to the order of h^2 .

21.24 Fourier's law is used routinely by architectural engineers to determine heat flow through walls. The following temperatures are measured from the surface ($x = 0$) into a stone wall:

$x, \text{ m}$	0	0.08	0.16
$T, ^\circ\text{C}$	19	17	15

If the flux at $x = 0$ is 60 W/m², compute k .

21.25 The horizontal surface area A_s (m²) of a lake at a particular depth can be computed from volume by differentiation:

$$A_s(z) = \frac{dV}{dz}(z)$$

where V = volume (m³) and z = depth (m) as measured from the surface down to the bottom. The average concentration of a substance that varies with depth, $\bar{c}$ (g/m³), can be computed by integration:

$$\bar{c} = \frac{\int_0^Z c(z) A_s(z) dz}{\int_0^Z A_s(z) dz}$$

where Z = the total depth (m). Determine the average concentration based on the following data:

$z, \text{ m}$	0	4	8	12	16
$V, 10^6 \text{ m}^3$	9.8175	5.1051	1.9635	0.3927	0.0000
$c, \text{ g/m}^3$	10.2	8.5	7.4	5.2	4.1

21.26 Faraday's law characterizes the voltage drop across an inductor as

$$V_L = L \frac{di}{dt}$$

where V_L = voltage drop (V), L = inductance (in henrys; 1 H = 1 V · s/A), i = current (A), and t = time (s). Determine the voltage drop as a function of time from the following data for an inductance of 4 H.

t	0	0.1	0.2	0.3	0.5	0.7
i	0	0.16	0.32	0.56	0.84	2.0

21.27 Based on Faraday's law (Prob. 21.26), use the following voltage data to estimate the inductance if a current of 2 A is passed through the inductor over 400 milliseconds.

$t, \text{ ms}$	0	10	20	40	60	80	120	180	280	400
$V, \text{ volts}$	0	18	29	44	49	46	35	26	15	7

21.28 The rate of cooling of a body (Fig. P21.28) can be expressed as

$$\frac{dT}{dt} = -k(T - T_a)$$

where T = temperature of the body (°C), T_a = temperature of the surrounding medium (°C), and k = a proportionality constant (per minute). Thus, this equation (called *Newton's law of cooling*) specifies that the rate of cooling is proportional to

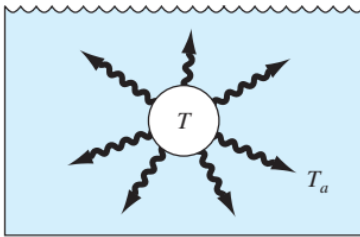


FIGURE P21.28

the difference in the temperatures of the body and of the surrounding medium. If a metal ball heated to 80 °C is dropped into water that is held constant at $T_a = 20$ °C, the temperature of the ball changes, as in

Time, min	0	5	10	15	20	25
$T, ^\circ\text{C}$	80	44.5	30.0	24.1	21.7	20.7

Utilize numerical differentiation to determine dT/dt at each value of time. Plot dT/dt versus $T - T_a$ and employ linear regression to evaluate k .

21.29 The enthalpy of a real gas is a function of pressure as described below. The data were taken for a real fluid. Estimate the enthalpy of the fluid at 400 K and 50 atm (evaluate the integral from 0.1 atm to 50 atm).

$$H = \int_0^P \left(V - T \left(\frac{\partial V}{\partial T} \right)_P \right) dP$$

P, atm	V, L		
	$T = 350 \text{ K}$	$T = 400 \text{ K}$	$T = 450 \text{ K}$
0.1	220	250	282.5
5	4.1	4.7	5.23
10	2.2	2.5	2.7
20	1.35	1.49	1.55
25	1.1	1.2	1.24
30	0.90	0.99	1.03
40	0.68	0.75	0.78
45	0.61	0.675	0.7
50	0.54	0.6	0.62

21.30 For fluid flow over a surface, the heat flux to the surface can be computed with Fourier's law: y = distance normal to the surface (m). The following measurements are

made for air flowing over a flat plate where y = distance normal to the surface:

y, cm	0	1	3	5
T, K	900	480	270	210

If the plate's dimensions are 200 cm long and 50 cm wide, and $k = 0.028 \text{ J/(s} \cdot \text{m} \cdot \text{K)}$, (a) determine the flux at the surface, and (b) the heat transfer in watts. Note that $1 \text{ J} = 1 \text{ W} \cdot \text{s}$.

21.31 The pressure gradient for laminar flow through a constant radius tube is given by

$$\frac{dp}{dx} = -\frac{8\mu Q}{\pi r^4}$$

where p = pressure (N/m^2), x = distance along the tube's centerline (m), μ = dynamic viscosity ($\text{N} \cdot \text{s/m}^2$), Q = flow (m^3/s) and r = radius (m).

(a) Determine the pressure drop for a 10-cm length tube for a viscous liquid ($\mu = 0.005 \text{ N} \cdot \text{s/m}^2$, density = $\rho = 1 \times 10^3 \text{ kg/m}^3$) with a flow of $10 \times 10^{-6} \text{ m}^3/\text{s}$ and the following varying radii along its length:

x, cm	0	2	4	5	6	7	10
r, mm	2	1.35	1.34	1.6	1.58	1.42	2

(b) Compare your result with the pressure drop that would have occurred if the tube had a constant radius equal to the average radius.

(c) Determine the average Reynolds number for the tube to verify that flow is truly laminar ($\text{Re} = \rho v D / \mu < 2100$ where v = velocity).

21.32 The following data for the specific heat of benzene were generated with an n th-order polynomial. Use numerical differentiation to determine n .

T, K	300	400	500	600
$C_p, \text{kJ/(kmol} \cdot \text{K)}$	82.888	112.136	136.933	157.744

T, K	700	800	900	1000
$C_p, \text{kJ/(kmol} \cdot \text{K)}$	175.036	189.273	200.923	210.450

21.33 The specific heat at constant pressure c_p [$\text{J/(kg} \cdot \text{K)}$] of an ideal gas is related to enthalpy by

$$c_p = \frac{dh}{dT}$$

where h = enthalpy (kJ/kg), and T = absolute temperature (K). The following enthalpies are provided for carbon

dioxide (CO_2) at several temperatures. Use these values to determine the specific heat in $\text{J}/(\text{kg} \cdot \text{K})$ for each of the tabulated temperatures. Note that the atomic weights of carbon and oxygen are 12.011 and 15.9994 g/mol, respectively

T, K	750	800	900	1000
$h, \text{kJ/kmol}$	29,629	32,179	37,405	42,769

21.34 An n th-order rate law is often used to model chemical reactions that solely depend on the concentration of a single reactant:

$$\frac{dc}{dt} = -kc^n$$

where c = concentration (mole), t = time (min), n = reaction order (dimensionless), and k = reaction rate ($\text{min}^{-1} \text{mole}^{1-n}$). The *differential method* can be used to evaluate the parameters k and n . This involves applying a logarithmic transform to the rate law to yield,

$$\log\left(-\frac{dc}{dt}\right) = \log k + n \log c$$

Therefore, if the n th-order rate law holds, a plot of the $\log(-dc/dt)$ versus $\log c$ should yield a straight line with a slope of n and an intercept of $\log k$. Use the differential method and linear regression to determine k and n for the following data for the conversion of ammonium cyanate to urea:

t, min	0	5	15	30	45
c, mole	0.750	0.594	0.420	0.291	0.223

21.35 The sediment oxygen demand [SOD in units of $\text{g}/(\text{m}^2 \cdot \text{d})$] is an important parameter in determining the dissolved oxygen content of a natural water. It is measured by placing a sediment core in a cylindrical container (Fig. P21.35). After carefully introducing a layer of distilled, oxygenated water above the sediments, the container is covered to prevent gas transfer. A stirrer is used to mix the water gently, and an oxygen probe tracks how the water's oxygen concentration decreases over time. The SOD can then be computed as

$$\text{SOD} = -H \frac{do}{dt}$$

where H = the depth of water (m), o = oxygen concentration (g/m^3), and t = time (d).

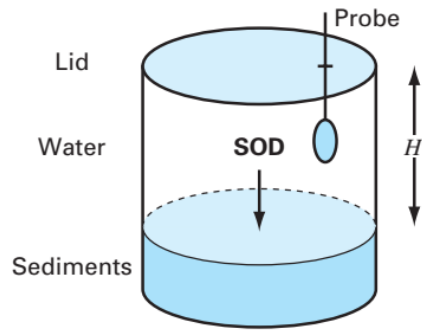


FIGURE P21.35

Based on the following data and $H = 0.1$ m, use numerical differentiation to generate plots of **(a)** SOD versus time and **(b)** SOD versus oxygen concentration:

t, d	0	0.125	0.25	0.375	0.5	0.625	0.75
$o, \text{mg/L}$	10	7.11	4.59	2.57	1.15	0.33	0.03

21.36 The following relationships can be used to analyze uniform beams subject to distributed loads:

$$\frac{dy}{dx} = \theta(x) \quad \frac{d\theta}{dx} = \frac{M(x)}{EI} \quad \frac{dM}{dx} = V(x) \quad \frac{dV}{dx} = -w(x)$$

where x = distance along beam (m), y = deflection (m), $\theta(x)$ = slope (m/m), E = modulus of elasticity ($\text{Pa} = \text{N}/\text{m}^2$), I = moment of inertia (m^4), $M(x)$ = moment (N m), $V(x)$ = shear (N), and $w(x)$ = distributed load (N/m). For the case of a linearly increasing load (recall Fig. P5.13), the slope can be computed analytically as

$$\theta(x) = \frac{w_0}{120EIL}(-5x^4 + 6L^2x^2 - L^4) \quad (\text{P21.36})$$

Employ **(a)** numerical integration to compute the deflection (in m) and **(b)** numerical differentiation to compute the moment (in N m) and shear (in N). Base your numerical calculations on values of the slope computed with Eq. P21.36 at equally spaced intervals of $\Delta x = 0.125$ m along a 3-m beam. Use the following parameter values in your computation: $E = 200$ GPa, $I = 0.0003$ m^4 , and $w_0 = 2.5$ kN/cm. In addition, the deflections at the ends of the beam are set at $y(0) = y(L) = 0$. Be careful of units.

21.37 You measure the following deflections along the length of a simply supported uniform beam (see Prob. 21.36)

x, m	0	0.375	0.75	1.125	1.5
y, cm	0	-0.2571	-0.9484	-1.9689	-3.2262
x, m	1.875	2.25	2.625	3	
y, cm	-4.6414	-6.1503	-7.7051	-9.275	

Employ numerical differentiation to compute the slope, the moment (in N m), the shear (in N) and the distributed load (in N/m). Use the following parameter values in your computation: $E = 200$ GPa, and $I = 0.0003$ m⁴.

21.38 Evaluate $\partial f/\partial x$, $\partial f/\partial y$, and $\partial f/(\partial x \partial y)$ for the following function at $x = y = 1$ (a) analytically and (b) numerically $\Delta x = \Delta y = 0.0001$:

$$f(x, y) = 3xy + 3x - x^3 - 3y^3$$

21.39 Develop a script to generate the same computations and plots as in Sec. 21.8, but for the following functions (for $x = -3$ to 3 and $y = -3$ to 3): (a) $f(x, y) = e^{-(x^2+y^2)}$ and (b) $f(x, y) = xe^{-(x^2+y^2)}$.

21.40 Develop a script to generate the same computations and plots as in Sec. 21.8, but for the MATLAB peaks function over ranges of both x and y from -3 to 3.

PROBLEMS

22.1 Solve the following initial value problem over the interval from $t = 0$ to 2 where $y(0) = 1$. Display all your results on the same graph.

$$\frac{dy}{dt} = yt^3 - 1.5y$$

- (a) Analytically.
- (b) Using Euler's method with $h = 0.5$ and 0.25 .
- (c) Using the midpoint method with $h = 0.5$.
- (d) Using the fourth-order RK method with $h = 0.5$.

22.2 Solve the following problem over the interval from $x = 0$ to 1 using a step size of 0.25 where $y(0) = 1$. Display all your results on the same graph.

$$\frac{dy}{dx} = (1 + 4x)\sqrt{y}$$

- (a) Analytically.
- (b) Using Euler's method.
- (c) Using Heun's method without iteration.
- (d) Using Ralston's method.
- (e) Using the fourth-order RK method.

22.3 Solve the following problem over the interval from $t = 0$ to 2 using a step size of 0.5 where $y(0) = 1$. Display all your results on the same graph.

$$\frac{dy}{dt} = -2y + t^2$$

Obtain your solutions with (a) Heun's method without iterating the corrector, (b) Heun's method with iterating the corrector until $\varepsilon_s < 0.1\%$, (c) the midpoint method, and (d) Ralston's method.

22.4 The growth of populations of organisms has many engineering and scientific applications. One of the simplest models assumes that the rate of change of the population p is proportional to the existing population at any time t :

$$\frac{dp}{dt} = k_g p \quad (\text{P22.4.1})$$

where k_g = the growth rate. The world population in millions from 1950 through 2000 was

t	1950	1955	1960	1965	1970	1975
p	2560	2780	3040	3350	3710	4090
t	1980	1985	1990	1995	2000	
p	4450	4850	5280	5690	6080	

- (a) Assuming that Eq. (P22.4.1) holds, use the data from 1950 through 1970 to estimate k_g .
 (b) Use the fourth-order RK method along with the results of (a) to stimulate the world population from 1950 to 2050 with a step size of 5 years. Display your simulation results along with the data on a plot.

22.5 Although the model in Prob. 22.4 works adequately when population growth is unlimited, it breaks down when factors such as food shortages, pollution, and lack of space inhibit growth. In such cases, the growth rate is not a constant, but can be formulated as

$$k_g = k_{gm}(1 - p/p_{\max})$$

where k_{gm} = the maximum growth rate under unlimited conditions, p = population, and $p_{\max}$ = the maximum population. Note that $p_{\max}$ is sometimes called the *carrying capacity*. Thus, at low population density $p \ll p_{\max}$, $k_g \rightarrow k_{gm}$. As p approaches $p_{\max}$, the growth rate approaches zero. Using this growth rate formulation, the rate of change of population can be modeled as

$$\frac{dp}{dt} = k_{gm}(1 - p/p_{\max})p$$

This is referred to as the *logistic model*. The analytical solution to this model is

$$p = p_0 \frac{p_{\max}}{p_0 + (p_{\max} - p_0)e^{-k_{gm}t}}$$

Simulate the world's population from 1950 to 2050 using (a) the analytical solution, and (b) the fourth-order RK method with a step size of 5 years. Employ the following initial conditions and parameter values for your simulation: p_0 (in 1950) = 2,560 million people, $k_{gm} = 0.026/\text{yr}$, and $p_{\max} = 12,000$ million people. Display your results as a plot along with the data from Prob. 22.4.

22.6 Suppose that a projectile is launched upward from the earth's surface. Assume that the only force acting on the object is the downward force of gravity. Under these conditions, a force balance can be used to derive

$$\frac{dv}{dt} = -g(0) \frac{R^2}{(R+x)^2}$$

where v = upward velocity (m/s), t = time (s), x = altitude (m) measured upward from the earth's surface, $g(0)$ = the gravitational acceleration at the earth's surface ($\cong 9.81 \text{ m/s}^2$), and R = the earth's radius ($\cong 6.37 \times 10^6 \text{ m}$). Recognizing that $dx/dt = v$, use Euler's method to determine the maximum height that would be obtained if $v(t=0) = 1500 \text{ m/s}$.

22.7 Solve the following pair of ODEs over the interval from $t = 0$ to 0.4 using a step size of 0.1. The initial conditions are $y(0) = 2$ and $z(0) = 4$. Obtain your solution with (a) Euler's method and (b) the fourth-order RK method. Display your results as a plot.

$$\begin{aligned} \frac{dy}{dt} &= -2y + 5e^{-t} \\ \frac{dz}{dt} &= -\frac{yz^2}{2} \end{aligned}$$

22.8 The *van der Pol equation* is a model of an electronic circuit that arose back in the days of vacuum tubes:

$$\frac{d^2y}{dt^2} - (1 - y^2) \frac{dy}{dt} + y = 0$$

Given the initial conditions, $y(0) = y'(0) = 1$, solve this equation from $t = 0$ to 10 using Euler's method with a step size of (a) 0.25 and (b) 0.125. Plot both solutions on the same graph.

22.9 Given the initial conditions, $y(0) = 1$ and $y'(0) = 0$, solve the following initial-value problem from $t = 0$ to 4:

$$\frac{d^2y}{dt^2} + 4y = 0$$

Obtain your solutions with (a) Euler's method and (b) the fourth-order RK method. In both cases, use a step size of 0.1. Plot both solutions on the same graph along with the exact solution $y = \cos 2t$.

22.10 Develop an M-file to solve a single ODE with Heun's method with iteration. Design the M-file so that it creates a plot of the results. Test your program by using it to solve for population as described in Prob. 22.5. Employ a step size of 5 years and iterate the corrector until $\varepsilon_s < 0.1\%$.

22.11 Develop an M-file to solve a single ODE with the midpoint method. Design the M-file so that it creates a plot of the results. Test your program by using it to solve for population as described in Prob. 22.5. Employ a step size of 5 years.

22.12 Develop an M-file to solve a single ODE with the fourth-order RK method. Design the M-file so that it creates a plot of the results. Test your program by using it to solve Prob. 22.2. Employ a step size of 0.1.

22.13 Develop an M-file to solve a system of ODEs with Euler's method. Design the M-file so that it creates a plot of the results. Test your program by using it to solve Prob. 22.7 with a step size of 0.25.

22.14 Isle Royale National Park is a 210-square-mile archipelago composed of a single large island and many small islands in Lake Superior. Moose arrived around 1900, and by 1930, their population approached 3000, ravaging vegetation. In 1949, wolves crossed an ice bridge from Ontario. Since the late 1950s, the numbers of the moose and wolves have been tracked.

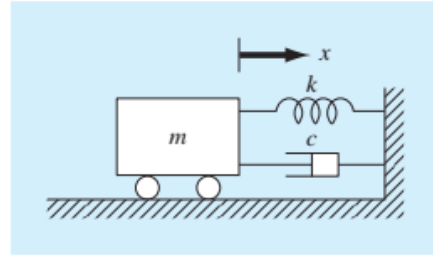


FIGURE P22.15

(a) Integrate the Lotka-Volterra equations (Sec. 22.6) from 1960 through 2020 using the following coefficient values: $a = 0.23$, $b = 0.0133$, $c = 0.4$, and $d = 0.0004$. Compare your simulation with the data using a time-series plot and determine the sum of the squares of the residuals between your model and the data for both the moose and the wolves.

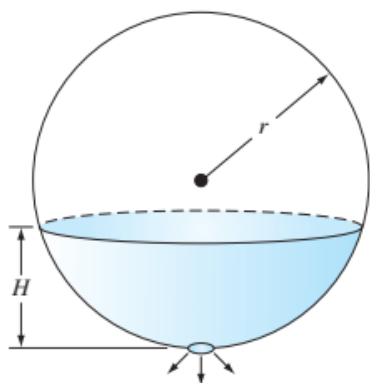
(b) Develop a phase-plane plot of your solution.

22.15 The motion of a damped spring-mass system (Fig. P22.15) is described by the following ordinary differential equation:

$$m \frac{d^2 x}{dt^2} + c \frac{dx}{dt} + kx = 0$$

where x = displacement from equilibrium position (m), t = time (s), $m = 20$ -kg mass, and c = the damping coefficient ($\text{N} \cdot \text{s/m}$). The damping coefficient c takes on three values

Year	Moose	Wolves	Year	Moose	Wolves	Year	Moose	Wolves
1959	563	20	1975	1355	41	1991	1313	12
1960	610	22	1976	1282	44	1992	1590	12
1961	628	22	1977	1143	34	1993	1879	13
1962	639	23	1978	1001	40	1994	1770	17
1963	663	20	1979	1028	43	1995	2422	16
1964	707	26	1980	910	50	1996	1163	22
1965	733	28	1981	863	30	1997	500	24
1966	765	26	1982	872	14	1998	699	14
1967	912	22	1983	932	23	1999	750	25
1968	1042	22	1984	1038	24	2000	850	29
1969	1268	17	1985	1115	22	2001	900	19
1970	1295	18	1986	1192	20	2002	1100	17
1971	1439	20	1987	1268	16	2003	900	19
1972	1493	23	1988	1335	12	2004	750	29
1973	1435	24	1989	1397	12	2005	540	30
1974	1467	31	1990	1216	15	2006	450	30

**FIGURE P22.16**

A spherical tank.

of 5 (underdamped), 40 (critically damped), and 200 (overdamped). The spring constant $k = 20$ N/m. The initial velocity is zero, and the initial displacement $x = 1$ m. Solve this equation using a numerical method over the time period $0 \leq t \leq 15$ s. Plot the displacement versus time for each of the three values of the damping coefficient on the same plot.

22.16 A spherical tank has a circular orifice in its bottom through which the liquid flows out (Fig. P22.16). The flow rate through the hole can be estimated as

$$Q_{\text{out}} = CA\sqrt{2gh}$$

where Q_{out} = outflow (m^3/s), C = an empirically derived coefficient, A = the area of the orifice (m^2), g = the gravitational constant ($= 9.81 \text{ m/s}^2$), and h = the depth of liquid in the tank. Use one of the numerical methods described in this chapter to determine how long it will take for the water to flow out of a 3-m diameter tank with an initial height of 2.75 m. Note that the orifice has a diameter of 3 cm and $C = 0.55$.

22.17 In the investigation of a homicide or accidental death, it is often important to estimate the time of death. From the experimental observations, it is known that the surface temperature of an object changes at a rate proportional to the difference between the temperature of the object and that of the surrounding environment or ambient temperature. This is known as Newton's law of cooling. Thus, if $T(t)$ is the temperature of the object at time t , and T_a is the constant ambient temperature:

$$\frac{dT}{dt} = -K(T - T_a)$$

where $K > 0$ is a constant of proportionality. Suppose that at time $t = 0$ a corpse is discovered and its temperature is measured to be T_o . We assume that at the time of death, the body temperature T_d was at the normal value of 37°C .

Suppose that the temperature of the corpse when it was discovered was 29.5°C , and that two hours later, it is 23.5°C . The ambient temperature is 20°C .

(a) Determine K and the time of death.

(b) Solve the ODE numerically and plot the results.

22.18 The reaction $A \rightarrow B$ takes place in two reactors in series. The reactors are well mixed but are not at steady state. The unsteady-state mass balance for each stirred tank reactor is shown below:

$$\frac{dCA_1}{dt} = \frac{1}{\tau}(CA_0 - CA_1) - kCA_1$$

$$\frac{dCB_1}{dt} = \frac{1}{\tau}CB_1 + kCA_1$$

$$\frac{dCA_2}{dt} = \frac{1}{\tau}(CA_1 - CA_2) - kCA_2$$

$$\frac{dCB_2}{dt} = \frac{1}{\tau}(CB_1 - CB_2) - kCB_2$$

where CA_0 = concentration of A at the inlet of the first reactor, CA_1 = concentration of A at the outlet of the first reactor (and inlet of the second), CA_2 = concentration of A at the outlet of the second reactor, CB_1 = concentration of B at the outlet of the first reactor (and inlet of the second), CB_2 = concentration of B in the second reactor, τ = residence time for each reactor, and k = the rate constant for reaction of A to produce B . If CA_0 is equal to 20, find the concentrations of A and B in both reactors during their first 10 minutes of operation. Use $k = 0.12/\text{min}$ and $\tau = 5$ min and assume that the initial conditions of all the dependent variables are zero.

22.19 A nonisothermal batch reactor can be described by the following equations:

$$\frac{dC}{dt} = -e^{(-10/(T+273))}C$$

$$\frac{dT}{dt} = 1000e^{(-10/(T+273))}C - 10(T - 20)$$

where C is the concentration of the reactant and T is the temperature of the reactor. Initially, the reactor is at 16°C and has a concentration of reactant C of 1.0 gmol/L . Find the concentration and temperature of the reactor as a function of time.

22.20 The following equation can be used to model the deflection of a sailboat mast subject to a wind force:

$$\frac{d^2y}{dz^2} = \frac{f(z)}{2EI}(L - z)^2$$

where $f(z)$ = wind force, E = modulus of elasticity, L = mast length, and I = moment of inertia. Note that the force

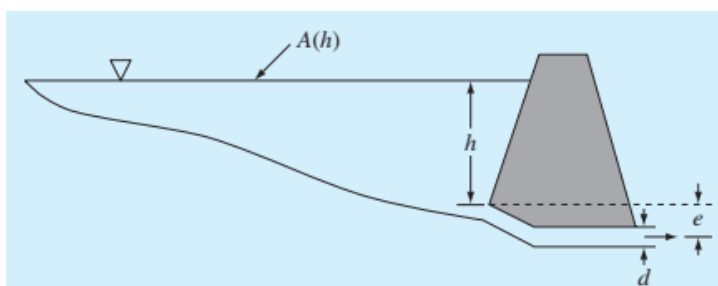


FIGURE P22.21

varies with height according to

$$f(z) = \frac{200z}{5+z} e^{-2z/30}$$

Calculate the deflection if $y = 0$ and $dy/dz = 0$ at $z = 0$. Use parameter values of $L = 30$, $E = 1.3 \times 10^8$, and $I = 0.05$ for your computation.

22.21 A pond drains through a pipe as shown in Fig. P22.21. Under a number of simplifying assumptions, the following differential equation describes how depth changes with time:

$$\frac{dh}{dt} = -\frac{\pi d^2}{4A(h)} \sqrt{2g(h+e)}$$

where h = depth (m), t = time (s), d = pipe diameter (m), $A(h)$ = pond surface area as a function of depth (m^2), g = gravitational constant ($= 9.81 \text{ m/s}^2$), and e = depth of pipe outlet below the pond bottom (m). Based on the following area-depth table, solve this differential equation to determine how long it takes for the pond to empty, given that $h(0) = 6 \text{ m}$, $d = 0.25 \text{ m}$, $e = 1 \text{ m}$.

$h, \text{ m}$	6	5	4	3	2	1	0
$A(h), \text{ m}^2$	1.17	0.97	0.67	0.45	0.32	0.18	0

22.22 Engineers and scientists use mass-spring models to gain insight into the dynamics of structures under the influence of disturbances such as earthquakes. Figure P22.22 shows such a representation for a three-story building. For this case, the analysis is limited to horizontal motion of the structure. Using Newton's second law, force balances can be developed for this system as

$$\frac{d^2 x_1}{dt^2} = -\frac{k_1}{m_1} x_1 + \frac{k_2}{m_1} (x_2 - x_1)$$

$$\frac{d^2 x_2}{dt^2} = \frac{k_2}{m_2} (x_1 - x_2) + \frac{k_3}{m_2} (x_3 - x_2)$$

$$\frac{d^2 x_3}{dt^2} = \frac{k_3}{m_3} (x_2 - x_3)$$

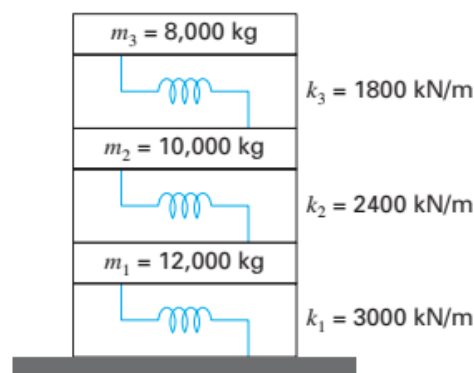


FIGURE P22.22

Simulate the dynamics of this structure from $t = 0$ to 20 s, given the initial condition that the velocity of the ground floor is $dx_1/dt = 1 \text{ m/s}$, and all other initial values of displacements and velocities are zero. Present your results as two time-series plots of (a) displacements and (b) velocities. In addition, develop a three-dimensional phase-plane plot of the displacements.

22.23 Repeat the the same simulations as in Section 22.6 for the Lorenz equations but generate the solutions with the midpoint method.

22.24 Perform the same simulations as in Section 22.6 for the Lorenz equations but use a value of $r = 99.96$. Compare your results with those obtained in Section 22.6.

where $w_o = 450$ N/m. The slope of the cable ($dy/dx = 0$) at $x = 0$, which is the lowest point for the cable. It is also the point where the tension in the cable is a minimum of T_o . The differential equation which governs the cable is

$$\frac{d^2 y}{dx^2} = \frac{w_o}{T_o} \left[1 + \sin \left(\frac{\pi x}{2l_A} \right) \right]$$

Solve this equation using a numerical method and plot the shape of the cable (y versus x). For the numerical solution, the value of T_o is unknown, so the solution must use an iterative technique, similar to the shooting method, to converge on a correct value of h_A for various values of T_o .

24.16 The basic differential equation of the elastic curve for a simply supported, uniformly loaded beam (Fig. P24.16) is given as

$$EI \frac{d^2 y}{dx^2} = \frac{wLx}{2} - \frac{wx^2}{2}$$

where E = the modulus of elasticity, and I = the moment of inertia. The boundary conditions are $y(0) = y(L) = 0$. Solve for the deflection of the beam using (a) the finite-difference approach ($\Delta x = 0.6$ m) and (b) the shooting method. The following parameter values apply: $E = 200$ GPa, $I = 30,000$ cm⁴, $w = 15$ kN/m, and $L = 3$ m. Compare your

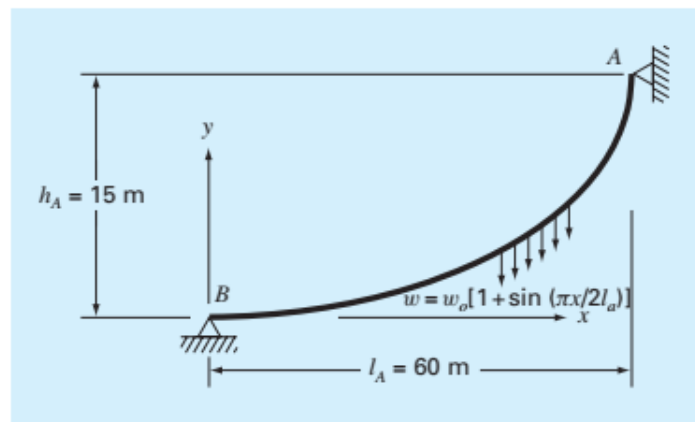
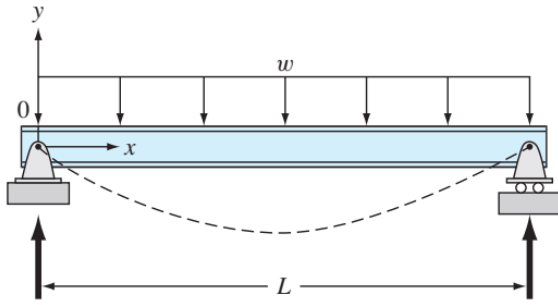


FIGURE P24.15

**FIGURE P24.16**

numerical results to the analytical solution:

$$y = \frac{wLx^3}{12EI} - \frac{wx^4}{24EI} - \frac{wL^3x}{24EI}$$

24.17 In Prob. 24.16, the basic differential equation of the elastic curve for a uniformly loaded beam was formulated as

$$EI \frac{d^2y}{dx^2} = \frac{wLx}{2} - \frac{wx^2}{2}$$

Note that the right-hand side represents the moment as a function of x . An equivalent approach can be formulated in terms of the fourth derivative of deflection as

$$EI \frac{d^4y}{dx^4} = -w$$

For this formulation, four boundary conditions are required. For the supports shown in Fig. P24.16, the conditions are

that the end displacements are zero, $y(0) = y(L) = 0$, and that the end moments are zero, $y''(0) = y''(L) = 0$. Solve for the deflection of the beam using the finite-difference approach ($\Delta x = 0.6$ m). The following parameter values apply: $E = 200$ GPa, $I = 30,000$ cm⁴, $w = 15$ kN/m, and $L = 3$ m. Compare your numerical results with the analytical solution given in Prob. 24.16.

24.18 Under a number of simplifying assumptions, the steady-state height of the water table in a one-dimensional, unconfined groundwater aquifer (Fig. P24.18) can be modeled with the following second-order ODE:

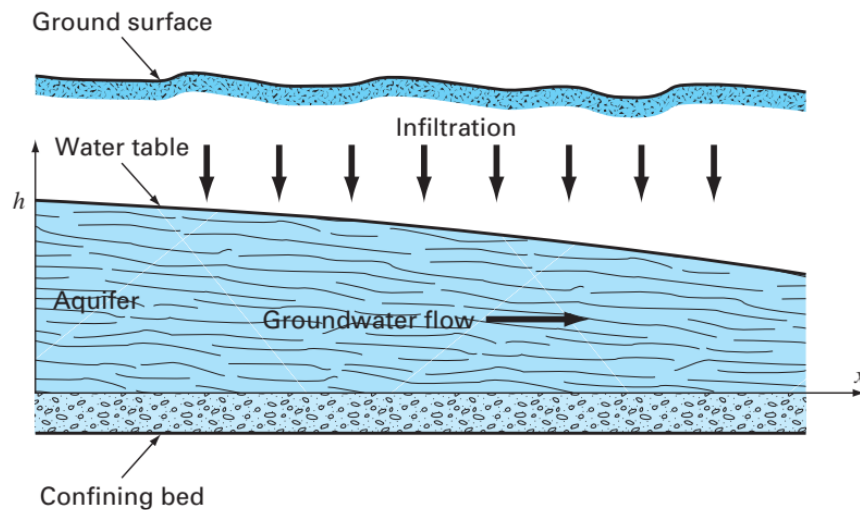
$$K\bar{h} \frac{d^2h}{dx^2} + N = 0$$

where x = distance (m), K = hydraulic conductivity (m/d), h = height of the water table (m), $\bar{h}$ = the average height of the water table (m), and N = infiltration rate (m/d).

Solve for the height of the water table for $x = 0$ to 1000 m where $h(0) = 10$ m and $h(1000) = 5$ m. Use the following parameters for the calculation: $K = 1$ m/d and $N = 0.0001$ m/d. Set the average height of the water table as the average of the boundary conditions. Obtain your solution with (a) the shooting method and (b) the finite-difference method ($\Delta x = 100$ m).

24.19 In Prob. 24.18, a linearized groundwater model was used to simulate the height of the water table for an unconfined aquifer. A more realistic result can be obtained by using the following nonlinear ODE:

$$\frac{d}{dx} \left(Kh \frac{dh}{dx} \right) + N = 0$$

**FIGURE P24.18**

An unconfined or "phreatic" aquifer.

where x = distance (m), K = hydraulic conductivity (m/d), h = height of the water table (m), and N = infiltration rate (m/d). Solve for the height of the water table for the same case as in Prob. 24.18. That is, solve from $x = 0$ to 1000 m with $h(0) = 10$ m, $h(1000) = 5$ m, $K = 1$ m/d, and $N = 0.0001$ m/d. Obtain your solution with (a) the shooting method and (b) the finite-difference method ($\Delta x = 100$ m).

24.20 Just as Fourier's law and the heat balance can be employed to characterize temperature distribution, analogous relationships are available to model field problems in other areas of engineering. For example, electrical engineers use a similar approach when modeling electrostatic fields. Under a number of simplifying assumptions, an analog of Fourier's law can be represented in one-dimensional form as

$$D = -\varepsilon \frac{dV}{dx}$$

where D is called the electric flux density vector, ε = permittivity of the material, and V = electrostatic potential. Similarly, a Poisson equation (see Prob. 24.8) for electrostatic fields can be represented in one dimension as

$$\frac{d^2V}{dx^2} = -\frac{\rho_v}{\varepsilon}$$

where ρ_v = charge density. Use the finite-difference technique with $\Delta x = 2$ to determine V for a wire where $V(0) = 1000$, $V(20) = 0$, $\varepsilon = 2$, $L = 20$, and $\rho_v = 30$.

24.21 Suppose that the position of a falling object is governed by the following differential equation:

$$\frac{d^2x}{dt^2} + \frac{c}{m} \frac{dx}{dt} - g = 0$$

where c = a first-order drag coefficient = 12.5 kg/s, m = mass = 70 kg, and g = gravitational acceleration = 9.81 m/s².

Use the shooting method to solve this equation for the boundary conditions:

$$\begin{aligned} x(0) &= 0 \\ x(12) &= 500 \end{aligned}$$

24.22 As in Fig. P24.22, an insulated metal rod has a fixed temperature (T_0) boundary condition at its left end. On its right end, it is joined to a thin-walled tube filled with water through which heat is conducted. The tube is insulated at its right end and convects heat with the surrounding fixed-temperature air (T_∞). The convective heat flux at a location x along the tube (W/m²) is represented by

$$J_{conv} = h(T_\infty - T_2(x))$$

where h = the convection heat transfer coefficient [W/(m² · K)]. Employ the finite-difference method with $\Delta x = 0.1$ m to compute the temperature distribution for the case where both the rod and tube are cylindrical with the same radius r (m). Use the following parameters for your analysis: $L_{rod} = 0.6$ m, $L_{tube} = 0.8$ m, $T_0 = 400$ K, $T_\infty = 300$ K, $r = 3$ cm, $\rho_1 = 7870$ kg/m³, $C_{p1} = 447$ J/(kg · K), $k_1 = 80.2$ W/(m · K), $\rho_2 = 1000$ kg/m³, $C_{p2} = 4.18$ kJ/(kg · K), $k_2 = 0.615$ W/(m · K), and $h = 3000$ W/(m² · K). The subscripts designate the rod (1) and the tube (2).

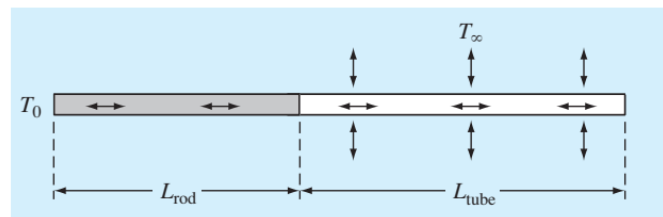


FIGURE P24.22

24.23 Perform the same calculation as in Prob. 24.22, but for the case where the tube is also insulated (i.e., no convection) and the right-hand wall is held at a fixed boundary temperature of 200 K.

Problems are taken From **Applied Numerical Methods by Chapra (3rd Edition)**